

UNIVERSITI POLY-TECH MALAYSIA

CYBERSAFE LEARNING HUB

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CYBERSAFE LEARNING HUB

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FYP4124 Page 4 of 263

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Abstract

The rapid expansion of digital technology has led to a significant increase in cybersecurity threats, yet awareness and understanding of these threats among university students remain critically low. Many students lack the foundational knowledge required to recognise and respond to common cyber risks such as phishing, malware, and weak password practices, making them vulnerable to attacks in both academic and personal digital environments. This project addresses this gap through the development of the CyberSafe Learning Hub, a web-based cybersecurity education platform designed to deliver structured, interactive, and gamified learning experiences to students.

The system was developed using the Laravel 12 framework with PHP 8.2, Livewire and Volt for reactive components, Bootstrap 5.3 for responsive design, and MySQL 8.0 as the database management system. The platform is deployed on a DigitalOcean cloud server using Docker containers managed through the Coolify self-hosting platform, ensuring reliable and scalable production performance.

Three core objectives guided the development of the system. The first objective was to improve cybersecurity awareness among students through structured lesson content. The second objective was to provide interactive and engaging education through multiple-choice quizzes with immediate feedback and an integrated AI-powered cybersecurity chatbot, built using the Groq API with the Meta Llama 3.3 70B Versatile model, which assists students with concept clarification during assessments. The third objective was to motivate students to apply cybersecurity knowledge through a gamification system featuring points, module-specific badges, a competitive leaderboard, and downloadable PDF certificates awarded upon category completion.

The system was evaluated through comprehensive testing covering non-functional, functional, and acceptance testing phases. All 18 non-functional test cases and 35 functional test cases across unit, integration, and system testing levels passed successfully. Client Acceptance Testing conducted with a lecturer at Kolej Teknologi Antarabangsa Cybernetics (KTAC) yielded a 100% acceptance rate across all 15 interview questions. User Acceptance Testing involving 54 student respondents recorded 100% positive responses across all evaluation criteria. The results confirm that the CyberSafe Learning Hub successfully meets all stated objectives and provides an effective, engaging, and secure platform for cybersecurity education.

Table of Contents

1 INTRODUCTION.....	17 1.1
Introduction	17 1.2
Project Background.....	17
1.3 Problem Statement.....	19
1.3.1 Low Cybersecurity Awareness Among Students	19
1.3.2 Limited Access to Interactive Cybersecurity Education.....	19
1.3.3 Lack of Motivation to Practice Cybersecurity Skills	19
1.4 Project Objectives	19
1.4.1 To Improve Cybersecurity Awareness Among Students	20
1.4.2 To Provide Interactive and Engaging Cybersecurity Education.....	20
1.4.3 To Motivate Students to Apply Cybersecurity Skills in Practice.....	20
1.5 Scope and Target User.....	20
1.5.1 Project Scope	20
1.5.2 Product Scope	21
1.5.3 Target User.....	21
1.6 Overview of This Report.....	22 2
LITERATURE REVIEW.....	23 2.1
Introduction	23
2.2 Investigation	23
2.2.1 Gamification in Cybersecurity Awareness	24
2.2.2 Mechanism of Gamified Learning.....	25
2.2.3 Case Studies of Gamification in Higher Education	26
2.2.4 Existing Tools and Platforms.....	26
2.2.5 Challenges and Limitations of Gamification in Cybersecurity Awareness.....	28
2.2.6 Investigation Summary Table.....	30
2.3 Related Works.....	31

2.3.1 TryHackMe	31
2.3.2 Cybrary.....	34
2.3.3 HackTheBox Academy	38
2.4 Comparison	41
2.5 Discussion	42
2.6 Conclusion.....	43
3 METHODOLOGY.....	44 3.1
Introduction	44 3.2
Agile Methodology.....	44
3.3 Phases in Agile Methodology	45
3.3.1 Phase 1: Planning.....	45
3.3.2 Phase 2: Design	46
3.3.3 Phase 3: Development.....	46
3.3.4 Phase 4: Testing.....	46
3.3.5 Phase 5: Deployment and Review	46
3.4 Requirement.....	47
FYP4124 Page 6 of 263	
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB	
3.4.1 Data Gathering Techniques	47 3.4.1.1
Questionnaire.....	47 3.4.1.2
Interview.....	47
3.4.2 Functional Requirement.....	48
3.4.3 Non-Functional Requirement	49
3.4.4 System Requirement	49
3.4.4.1 Hardware Requirement	49
3.4.4.2 Software Requirement.....	51
3.5 Analysis.....	54 3.5.1
Gathering Analysis	54 3.5.1.1
Analysis.....	55 3.5.1.2
Analysis.....	66 3.5.2
.....	69 3.5.3
Flowchart.....	71 3.5.3.1
Flowchart.....	71 3.5.3.2
Flowchart.....	73
3.6 Conclusion.....	74 4
DESIGN & IMPLEMENTATION.....	75 4.1
Introduction	75
4.2 Interface Design.....	75
4.2.1 Admin	77
4.2.2 Students	81
4.2.3 Mobile View (Student).....	85
4.3 Database Design.....	91
4.3.1 Data Dictionary	91
4.3.2 Data Flow Diagram (DFD)	104
4.3.3 Entity Relational Diagram (ERD).....	105
4.4 Security System Framework.....	107
4.4.1 Security Framework Diagram.....	107
4.5 Implementation Introduction	108
4.6 Execution Platform.....	108

4.6.1 Development Platform	109
4.6.2 Hosting Platform	110
4.7 Implementation Tools.....	111
4.7.1 Software	111
4.7.1.1 Visual Studio	111
4.7.1.2 Laravel	112
4.7.1.3 Livewire	112
4.7.1.4 Bootstrap	112
4.7.1.5 MySQL 8.0	113
4.7.1.6	113
4.7.1.7	113
4.7.1.8	114
4.7.1.9 Coolify	115
4.7.1.10	116
4.7.1.11	116
4.7.1.12 Groq	117
4.7.1.13	117
4.7.1.14 Composer.....	118
4.7.1.15 Github.....	119
4.7.2 Programming Language	120
4.7.2.1 PHP.....	120
4.7.2.2 JavaScript	121
4.7.2.3 HTML5	122
4.7.2.4 CSS.....	122
4.8 System Interface.....	123
4.8.1 Landing Page (Public).....	123
4.8.2 Login Page	124
4.8.3 Register Page.....	125
4.8.4 Google OAuth Login	126
4.8.5 Student Dashboard.....	127
4.8.6 Module Listing (Student View)	128
4.8.7 Module Detail with Lesson List (Student View).....	129
4.8.8 Lesson Content View (Student View).....	130
4.8.9 Quiz Interface (Student View)	131
4.8.10 Quiz Result.....	132
4.8.11 AI Chatbot (Quiz Page).....	133
4.8.12 Badges Page (Student View).....	134
4.8.13 Certificates Page (Student View)	135
4.8.14 Certificate Detail View (Student View).....	136
4.8.15 User Manual (Student View)	137
4.8.16 Admin Dashboard.....	138
4.8.17 Admin Category Management	139
4.8.18 Admin Create / Edit Category	140
4.8.19 Admin Module Management	140
4.8.20 Admin Create / Edit Module	141
4.8.21 Admin Lesson Management	142
4.8.22 Admin Lesson Content Editor	142
4.8.23 Admin Quiz Management	143
4.8.24 Admin Quiz Question Management	144
4.8.25 Admin Analytics Dashboard.....	144
4.8.26 Admin Student Analytics Drill-Down.....	145
4.8.27 Admin User Management	145
4.9 Significant Functions	146
4.9.1	146

Significant Functions.....	146	4.9.1.1 Google OAuth 2.0 Authentication	147	4.9.1.2 Quiz Scoring and Validation	148	4.9.1.3 AI Chatbot Integration.....	148
Elements.....	149	4.9.2 Security Control (RBAC) Middleware.....	149	4.9.2.2 Password Hashing and Data Protection	150	4.9.2.3 CSRF Protection	150
Access.....	151	4.9.2.4 Owner-Only Resource	151				

4.10 Conclusion152 5

FINDING..... 153 5.1

Introduction153 5.2

Testing.....153

FYP4124 Page 8 of 263

MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

5.3 Non-Functional Testing.....155

5.3.1 Performance Testing.....155

5.3.2 Security Testing.....156

5.3.3 Usability Testing157

5.3.4 Reliability Testing.....158

5.4 Functional Testing.....158

5.4.1 Unit Testing158

5.4.2 Integration Testing160

5.4.3 System Testing.....162

5.5 Acceptance Testing.....164

5.5.1 Client Acceptance Testing164

5.5.2 User Acceptance Testing.....168

5.6 Conclusion.....178 6

CONCLUSION 179 6.1

Introduction179

6.2 Project Schedule179

6.2.1 Work Breakdown Structure180

6.2.2 Gantt Chart.....180

6.3 Risk Management.....183

6.4 Achievement.....185

6.4.1 To Improve Cybersecurity Awareness Among Students185

6.4.2 To Provide Interactive and Engaging Cybersecurity Education186

6.4.3 To Motivate Students to Apply Cybersecurity Skills in Practice187

6.5 Constraint and Limitation188

6.5.1 Limited Cybersecurity Topic Coverage188

6.5.2 No Real-Time Collaboration or Discussion.....188

6.5.3 English-Only Content.....188

6.5.4 Manual Content Management.....189

6.5.5 Dependency on Third-Party AI API189

6.6 Future Work and Recommendation189

6.6.1 Expand Cybersecurity Content189

6.6.2 Multilingual Support189

6.6.3 Live Discussion and Peer Interaction.....190

6.6.4 Adaptive Learning Pathways.....190

6.6.5 Bulk Content Import	190
6.6.6 Mobile Application.....	190
6.6.7 Advanced Analytics and Reporting	190
6.6.8 Gamification Enhancements	191
6.7 Conclusion.....	191
Appendix A – Questionnaire & Interview Questions	192
Appendix B – User Manual.....	203
Appendix C – Turnitin Result.....	250
Appendix D – Log Book	256
References	262

Figure 2.1: Example of gamified awareness module interface (Razack & Saad, 2024).....	24
Figure 2.2: TryHackMe interface with progress badges.....	27
Figure 2.3: HackTheBox Academy challenge map.....	27
Figure 2.4: Cybrary learning roadmap dashboard.....	28
Figure 2.5: TryHackMe Dashboard.....	31
Figure 2.6: TryHackMe Learning Path Page.....	32
Figure 2.7: TryHackMe Challenge Room Interface.....	32
Figure 2.8: TryHackMe Leaderboard Interface.....	33
Figure 2.9: TryHackMe Completed Room Example.....	33
Figure 2.10: Cybrary Dashboard Page.....	35
Figure 2.11: Cybrary Course Catalog and Career Paths.....	35
Figure 2.12: Cybrary Course Module Page.....	36
Figure 2.13: Cybrary Learning Progress Dashboard.....	36
Figure 2.14: Cybrary Certification Preparation Page.....	37
Figure 2.15: HackTheBox Academy Dashboard Page.....	38
Figure 2.16: HackTheBox Academy Lab Interface Page.....	38
Figure 2.17: HackTheBox Academy Completion Badge Page.....	39
Figure 2.18: HackTheBox Academy Certificate Page.....	39
Figure 2.19: HackTheBox Academy Exams Page.....	40
Figure 3.1: Agile Methodology Diagram (Popa et al., 2021).....	45
Figure 3.2: Lenovo IdeaPad Gaming.....	50
Figure 3.3: Visual Studio Code Logo.....	51
Figure 3.4: Laravel 12 Logo.....	51
Figure 3.5: MySQL Logo.....	52
Figure 3.6: Bootstrap Logo.....	52
Figure 3.7: Google OAuth Logo.....	53
Figure 3.8: PHPUnit Logo.....	53
Figure 3.9: XAMPP Logo.....	54
Figure 3.10: Age of Respondents.....	56
Figure 3.11: Gender of Respondents.....	56
Figure 3.12: Program of Study.....	57
Figure 3.13: Frequency of Internet Use.....	57
Figure 3.14: Level of Cybersecurity Knowledge.....	58
Figure 3.15: Experience of Cybersecurity Incidents.....	58
Figure 3.16: Participation in Cybersecurity Awareness Program.....	59
Figure 3.17: Importance of Cybersecurity Awareness.....	59
Figure 3.18: Confidence in Protecting Personal Data.....	60
Figure 3.19: Preferred Learning Methods for Cybersecurity.....	60
Figure 3.20: Frequency of Applying Safe Online Practices.....	61
Figure 3.21: Awareness of Security Practices.....	61
Figure 3.22: Interest in Using Cybersecurity Learning Platforms.....	62
Figure 3.23: Importance of User Authentication.....	62
Figure 3.24: Motivation Through Gamification.....	63
Figure 3.25: Preferred Features in the CyberSafe Learning Hub.....	63
Figure 3.26: Usefulness of Progress Tracking.....	64
Figure 3.27: Importance of Platform Loading Speed.....	64
Figure 3.28: System Usability and Interface Design.....	65
Figure 3.29: Importance of Data Security and Encryption.....	65
Figure 3.30: Interview session with Mr.Syaheedullah.....	66

Figure 3.31: Use Case Diagram for CyberSafe Learning Hub	69
Figure 3.32: User (Student) Flowchart of CyberSafe Learning Hub.....	71
Figure 3.33: User (Administrator/Lecturer) Flowchart of CyberSafe Learning Hub.....	73
Figure 4.1: Landing Page.....	75
Figure 4.2: Login Page.....	76
Figure 4.3: Register Page.....	76
Figure 4.4: Admin Dashboard Page.....	77
Figure 4.5: Manage Module Page.....	78
Figure 4.6: Manage Lessons Page	78
Figure 4.7: Manage Quizzes Page	79
Figure 4.8: Student Analytics Page.....	79
Figure 4.9: Manage Users Page	80
Figure 4.10: Student Dashboard Page.....	81
Figure 4.11: View Module Page.....	82
Figure 4.12: View Lesson Page	82
Figure 4.13: Lesson Learning Page.....	83
Figure 4.14: Quiz Page.....	83
Figure 4.15: Quiz Result Page.....	84
Figure 4.16: Student Dashboard Page (Mobile View)	85
Figure 4.17: View Module Page (Mobile View)	86
Figure 4.18: View Lesson Page (Mobile View)	87
Figure 4.19: Lesson Learning Page (Mobile View)	88
Figure 4.20: Quiz Page (Mobile View)	89
Figure 4.21: Quiz Result Page (Mobile View)	90
Figure 4.22: Data Flow Diagram (DFD)	104
Figure 4.23: Entity Relationship Diagram (ERD).....	105
Figure 4.24: Security Framework Diagram (SFD)	107
Windows 11 Pro.....	109
Ubuntu 24.04 LTS	110
Visual Studio Code Interface	111
Docker Interface	114
DigitalOcean Interface	115
Coolify Interface	116
Groq AI – Llama 3.3 Interface	117
phpMyAdmin Interface	118
GitHub Interface	119
PHP Code Snippet – Quiz Result Controller (QuizAttemptController.php)	120
JavaScript Code Snippet – Quiz Timer (show.blade.php)	121
HTML5 Code Snippet – Quiz Question Template (show.blade.php)	122
CSS Code Snippet – Chat Widget Styling (show.blade.php)	123
CyberSafe Learning Hub Landing Page (Desktop View)	123
CyberSafe Learning Hub Landing Page (Mobile View)	124
Figure 4.40: Login Page (Desktop View)	124
Figure 4.41: Login Page (Mobile View)	125
Figure 4.42: Register Page (Desktop View)	125
Figure 4.43: Register Page (Mobile View)	126
Figure 4.44: Google OAuth 2.0 Authentication Screen (Desktop View)	126
Figure 4.45: Google OAuth 2.0 Authentication Screen (Mobile View)	127
Figure 4.46: Student Dashboard (Desktop View)	127
Figure 4.47: Student Dashboard (Mobile View)	128
Figure 4.48: Student Module Listing Page (Desktop View)	128
Figure 4.49: Student Module Listing Page (Mobile View)	129

Figure 4.50: Module Detail Page with Lesson List (Desktop View).....	129
Figure 4.51: Module Detail Page with Lesson List (Mobile View)	130
Figure 4.52: Lesson Content View (Desktop View)	130
Figure 4.53: Lesson Content View (Mobile View)	131
Figure 4.54: Quiz Interface (Desktop View)	131
Figure 4.55: Quiz Interface (Mobile View)	132
Figure 4.56: Quiz Result Page (Desktop View).....	132
Figure 4.57: Quiz Result Page (Mobile View)	133
Figure 4.58: AI Chatbot – Cybersecurity Tutor Assistant (Desktop View)	133
Figure 4.59: AI Chatbot – Cybersecurity Tutor Assistant (Mobile View)	134
Figure 4.60: Student Badges Page (Desktop View)	134
Figure 4.61: Student Badges Page (Mobile View)	135
Figure 4.62: Student Certificates Page (Desktop View)	135
Figure 4.63: Student Certificates Page (Mobile View)	136
Figure 4.64: Certificate Detail View (Desktop View)	136
Figure 4.65: Certificate Detail View (Mobile View)	137
Figure 4.66: Student User Manual Page (Desktop View)	137
Figure 4.67: Student User Manual Page (Mobile View)	138
Figure 4.68: Admin Dashboard	138
Figure 4.69: Admin Category Management	139
Figure 4.70: Admin Category Create / Edit Form	140
Figure 4.71: Admin Module Management	140
Figure 4.72: Admin Module Create / Edit Form	141
Figure 4.73: Admin Lesson Management	142
Figure 4.74: Admin Lesson Content Editor	142
Figure 4.75: Admin Quiz Management	143
Figure 4.76: Admin Quiz Detail and Question Management	144
Figure 4.77: Admin Analytics Dashboard	144
Figure 4.78: Admin Student Analytics – Individual Student View	145
Figure 4.79: Admin User Management	145
Figure 4.80: Code Snippet – Google OAuth 2.0 Callback (GoogleController.php)	147
Figure 4.81: Code Snippet – Quiz Submission and Validation (QuizAttemptController.php) ..	148
Figure 4.82: Code Snippet – AI Chatbot System Prompt (ChatbotController.php)	148
Figure 4.83: Code Snippet – Role-Based Access Control Middleware (RoleMiddleware.php) ..	149
Figure 4.84: Code Snippet – Password Hashing and Hidden Fields (User.php)	150
Figure 4.85: Code Snippet – CSRF Protection in Blade Form (show.blade.php)	150
Figure 4.86: Code Snippet – Owner-Only Access Control (QuizAttemptController.php)	151
Figure 5.1: Client Acceptance Testing Session with Mr.Syaheedullah bin Yahya	165
Figure 5.2: Q1 – Age Range of Respondents	168
Figure 5.3: Q2 – Gender of Respondents	169
Figure 5.4: Q3 – Registration & Login	169
Figure 5.5: Q4 – Navigation & Usability	170
Figure 5.6: Q5 – Cybersecurity Awareness (Lesson Structure)	170
Figure 5.7: Q6 – Cybersecurity Awareness (Content Clarity)	171
Figure 5.8: Q7 – Cybersecurity Awareness (Learning Outcome)	171
Figure 5.9: Q8 – Interactive Education (Quiz Design)	172
Figure 5.10: Q9 – Interactive Education (Visual Elements)	173
Figure 5.11: Q10 – Interactive Education (Quiz Feedback)	173
Figure 5.12: Q11 – Motivation & Gamification (Points System)	174
Figure 5.13: Q12 – Motivation & Gamification (Badge System)	174
Figure 5.14: Q13 – Motivation & Gamification (Progress Tracking)	175
Figure 5.15: Q14 – Functional Requirements	175

(Lesson & Quiz Access) 175

FYP4124 Page 13 of 263

MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

Figure 5.16: Q15 – Functional Requirements (Progress Recording)	176
Figure 5.17: Q16 – Non-Functional Requirements (Performance).....	177
Figure 5.18: Q17 – Non-Functional Requirements (Security)	177
Figure 6.1: Work Breakdown Structure of the CyberSafe Learning Hub Project	180
Figure 6.2: Gantt Chart of the CyberSafe Learning Hub Project Timeline	181

List of Tables

Table 2.1: Investigation Summary Table	30
Table 2.2: Comparison of Existing Project	41
Table 3.1: Functional Requirement Table	48
Table 3.2: Non-Functional Requirements Table	49

Table 3.3: Hardware Requirements Table	50
Table 3.4: Analysis of Interview Questions	67
Table 4.1: Users Table.....	91
Table 4.2: Roles Table.....	92
Table 4.3: User Roles (Pivot) Table	92
Table 4.4: Categories Table	93
Table 4.5: Modules Table	93
Table 4.6: Lessons Table.....	94
Table 4.7: Lesson Contents Table	95
Table 4.8: Lesson Comments Table	96
Table 4.9: Quizzes Table.....	96
Table 4.10: Questions Table.....	97
Table 4.11: Answers Table	97
Table 4.12: Quiz Attempts Table	98
Table 4.13: Quiz Answers Table.....	99
Table 4.14: Student Progress Table	99
Table 4.15: Badges Table.....	100
Table 4.16: User Badges Table	101
Table 4.17: Certificates Table	101
Table 4.18: User Certificates Table	102
Table 4.19: Achievement Notifications Table	103
Table 5.1: Testing Overview	154
Table 5.2: Performance Testing.....	155
Table 5.3: Security Testing	156
Table 5.4: Usability Testing.....	157
Table 5.5: Reliability Testing.....	158
Table 5.6: Unit Testing.....	159
Table 5.7: Integration Testing	160
Table 5.8: System Testing	162
Table 5.9: Client Acceptance Testing – Interview Response Summary.....	165
Table 6.1: Project Task Timeline and Duration	181
Table 6.2: Project Risk Management.....	183

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1 INTRODUCTION

1.1 Introduction

This chapter concludes the project by restating the project history and reasons for project initiation, the problem statement which identifies the critical elements of the problem which requires such a project, the project objectives and the project scope and target user. The chapter concludes with a brief overview of the rest of the report.

1.2 Project Background

In the 21st century, cybersecurity is a concern; digital technologies in education, work, and daily life have deeply rooted themselves. Among the heaviest users of these digital platforms are students engaging in online learning management systems, social media, cloud storage, and communication tools daily. However, this frequent usage does not equate to safe usage. Most students either lack awareness or discipline to follow cybersecurity best practices and thus expose themselves to cyber threats like phishing attacks, malware infection, poor password management, and unsafe online behaviors. According to studies conducted globally, young internet users often remain very confident about their digital skills, which contributes to their highly vulnerable position regarding cyber threats compared to more experienced users (Gabra et al., 2020).

This task is made more difficult because traditional cybersecurity training has fundamental limitations. Many awareness campaigns or lecture-based classes can impart theory, but many of these efforts are “dry as cardboard” and don’t engage students to change behaviours on a long-term basis. For instance, students know that they should use strong passwords but they reuse weak passwords because of their instant recall. This gap of knowing and doing exemplifies the necessity for education about cybersecurity awareness to be more interactive and engaging.

There are a few platforms out there that do this for CyberSafe Learning Hub, like TryHackMe, Cybrary and HackTheBox Academy. These resources range from virtual labs, to professional courses, to real-world simulations. But their major audience is IT undergraduate, working professionals or certification candidates. Consequently, the content can be extremely technical and expected former networking, programming or penetration testing tool knowledge. Though these platforms are great in professional progress, they do not cater for beginners or even non-IT students who need simple and easy-to-read and

age-appropriate resources. This opens a hole in the availability of security education tools, especially for IT-non related students in higher education.

Taking cognizance of this problem, Kolej Teknologi Antarabangsa Cybernetics (KTAC), a private institution of higher learning in Malaysia, has realised the need to incorporate cybersecurity awareness within its educational curriculum. KTAC realizes that cybersecurity learning is applicable not only to IT majors. All students across disciplines have interactions with digital systems such as turning in assignments, online banking and social networking and communication. With insufficient test and prepare, if their standoffishness throws them into phishing scams, thieving, identity theft, and other crimes in

cyberspace.

The problem that motivated this project was the low level of awareness and few easily accessible tools for dealing with cyber security in colleges. Workshops or seminars, when held at all, are typically a onetime activity and provide no guarantee that any learning will be retained or behaviour modified. Thus, KTAC needs a sustainable and organized solution, which easily integrates with the learning experiences of students, thus making them to be involved in cybersecurity awareness activities rather than being mere recipients of information.

In order to fill this gap, the proposed project will develop CyberSafe Learning Hub, a web application designed for KTAC students. The platform will deliver lessons with interactive quizzes and gamified challenges on key cybersecurity topics like phishing awareness, password safety, and safe browsing. Through gamifying the content through points, badges and progression, student have been more motivated to complete the course and students should not only learn this key stuff but learn how it is applied in their digital lives. Lecturers will be able to take advantage of an academic admin dashboard which allows content creation and management, the ability to track a student's performance over time. Students are at the centre and lecturers in support, two beneficial aspects of the platform that will make it last within an educational setting.

Ultimately, this project intends to fill the void between pro-level cyber security training environments out there and the lack of beginner-friendly college-oriented systems. Through a focus on accessibility, interactivity and motivation, the CyberSafe Learning Hub will enable KTAC students to develop safer online behaviours, while leaving the College with a trusted long-term method for educating young people about digital literacy safely navigating threats.

1.3 Problem Statement

Students in general also continue to struggle with awareness of cyber security. Despite regular use of digital devices/tablets and online platforms, the vast majority of students fail to comprehend the risks involved or what actions can safeguard them. Current awareness campaigns are often too mechanistic; they warrant already a bit of understanding but not being yet so formalized and probably still more making people aware. This thereby results in a requirement for an environment that would make cybersecurity to be presented in less formal, interactive and more motivational way especially where such learners are not IT inclined.

1.3.1 Low Cybersecurity Awareness Among Students

In general, and particularly in Malaysia, students (university and colleges) are often not equipped with appropriate knowledge about cybersecurity threats like phishing, malware spread, unsafe online behavior which could be exploited. For example, studies involving Malaysian university students report similar findings, where many lack essential cybersecurity practices and rely heavily on risky online habits (Gabra et al., 2020).

1.3.2 Limited Access to Interactive Cybersecurity Education

College-based conventional awareness means tend to be boring and/or students just did not get it, resulting in low involvement or limited knowledge retained. Studies have found that passive methods of instruction lead to only a modest change in behavior, but the impact can be enhanced by interactive and experiential training approach on enhancing internet users' awareness (Ng & Kamsin, 2021).

1.3.3 Lack of Motivation to Practice Cybersecurity Skills

Without gamified or hands-on experiences, students are less likely to apply cybersecurity knowledge in real life, leaving them vulnerable to online threats. Research demonstrates that gamification methods (e.g., quizzes, badges and leader boards), notably increase motivation and engagement in cyber security curricula (Gwenhure & Rahayu, 2024).

1.4 Project Objectives

The primary focus of the project is to design a system that engages college students for improved cyber security awareness education. Rather than passive learning, the device incorporates interactive and gamified learning for improved knowledge retention and motivation.

1.4.1 To Improve Cybersecurity Awareness Among Students

This goal is about increasing students' awareness of basic cyber threats like phishing, malware and weak passwords. Through organized, age-specific coursework, students will not only gain an understanding of what these dangers are, but also how to be safe in the digital world. This recommendation will better prepare students to defend themselves on the virtual landscape, mind the gap between their knowledge and the level they need to know.

1.4.2 To Provide Interactive and Engaging Cybersecurity Education

This goal highlights the requirement to achieve more than rote learning, a teaching method of dubious influence on knowledge retention among students. The platform will include interactive

elements such as brief quizzes, multiple-choice questions and visual collateral like infographics to foster an engaging learning process. By delivering straightforward, interactive lessons, students are engaged and more likely to pay attention, find the information useful, and use it in their digital lives.

1.4.3 To Motivate Students to Apply Cybersecurity Skills in Practice

This purpose is intended to engross students in learning cybersecurity and not give up through gamification. Students earn points by completing lessons and quizzes, see their progress represented in a nice infographic, and win badges for hitting milestones. It is through these simplistic elements of gamification that learning becomes a reward and fun, thereby promoting the continued use of learned material in daily digital activities.

1.5 Scope and Target User

This part of the project defines the scope and who will be affected by its use. By making the scope explicit, it helps to keep the project on track with what we can actually accomplish and prevents a reasonable idea from getting out of control. And if the target audience is determined, then it's clear for whom and how the system has been created.

1.5.1 Project Scope

The CyberSafe Learning Hub project is an initiative to develop a web-based platform that educates university students about key areas of cybersecurity awareness, including phishing, password security, and secure browsing. The platform will include organized modules, quizzes and gamification elements such as points, badges and progress to increase engagement, while a lecturer dashboard will facilitate lesson management and student participation.

FYP4124 Page 20 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

The system will be designed in Laravel 12 framework for the backend and MySQL database while Bootstrap for responsive frontend. By the end of this article, we will demonstrate how Google OAuth API can be used for secure login and later we use PHPUnit to test if the components work as intended- that is any new update should not lead to bugs. Despite relying on proven tools like Laravel, MySQL, and Bootstrap, more work must be poured into mastering the more intricate features of Laravel; into including gamification features in the application; as well as adapting leading content from other trusted sources like NCSA, OWASP and scholarly journals. The purpose is strictly educational and awareness raising, without really going deep into advanced penetration testing or cyber security learning at a pro level.

1.5.2 Product Scope

The product vision underscores functional as well as non-functional elements of the CyberSafe Learning Hub. In terms of functionality, the platform provides simplified cyber security knowledge by the way of interactive lessons, quizzes and playful content on safe internet usage, password security and digital responsibility. Gamification elements like gamified points, badges and progress will be used to make it more engaging. A lecturer dashboard is also offered so that the lecturers can design lessons, allocate modules and check students' performance for utilizing learning outcomes.

Non-functional-wise, the system will focus in usability with a clean and intuitive interface that make easy to most students or lecturers to navigate around on the platform. Performance requirements to be able to run smoothly and respond quickly when loading lessons, quizzes, or dashboards even during normal use. Security concerns will be met with the use of secure login and handling of student data, while reliability will assure consistent system availability for classroom time and individual use. The platform will also assist in making accessibility available to ensure that there is inclusivity for a wide range of learners.

Though the platform does an all-round job in creating a general sense of awareness within cyber security, it doesn't have coverage for more advanced technical topics like pentesting, malware analysis or cert-based training. Instead, the emphasis is on developing an intelligent basis of cybersecurity in college student's everyday lives.

1.5.3 Target User

This project is aimed at users in Kolej Teknologi Antarabangsa Cybernetics; primarily college students, particularly non-IT students who are ignorant towards cybersecurity. These students are heavy users of digital devices, and network consumes hence making them susceptible to common cyber threats in the absence of proper education.

FYP4124 Page 21 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

The platform is designed to teach them good cybersecurity habits they can use during their day-to-day digital pursuits by serving up digestible, easy-to-understand content. Lecturers besides students as secondary users will have access to a dashboard too, where they can author and monitor information of learning objects in their course. This duality in users permits the system to adequately support both teaching practices and learning events.

1.6 Overview of This Report

Chapter 1: Introduction

This chapter presents the background, problem statement, objectives, scope and target users of CyberSafe Learning Hub project. It is the base of the research and clarifies the need to inculcate cyber security awareness among students at Kolej Teknologi Antarabangsa Cybernetics.

Chapter 2: Literature Review

This chapter surveys literature, platforms, and approaches on cybersecurity awareness. It reviews the potentials of solutions like TryHackMe, Cybrary, and HackTheBox Academy and brings out their advantages as well as drawbacks. The description is intended to explore some deficiencies in existing methodologies, justify a need for a simpler and more gamified learning platform for college students.

Chapter 3: Methodology

In this chapter, we describe the approach used to create the CyberSafe Learning Hub. It teaches collecting, analyzing and translating user requirements to the system functionality. Surveys and interviews were used as data collection methods for students and lecturers to recognize user needs and expectations. The results from the study were interpreted to specify both functional and non-functional requirements for the system, on which the design of the system was based. Also, it complies requirements analysis and visual models like use case diagram and flow diagram to provide the understanding of how the system operates to serve its purpose.

2 LITERATURE REVIEW

2.1 Introduction

A literature review is a critical analysis of the studies, theories and practices conducted or used for the study area concerned. Its goal is to position the present project in a wider research context through identification of relevant pieces of work, description of main findings and gaps that have not been studied. In this way, the review provides a solid basis for the proposed investigation and shows how it relates to other scholarly conversations. Literature studies on online cyber security awareness, education gamification and existing systems or efforts to address them are conducted for this study. This section is dedicated to consolidating the literature and research projects in order to provide sound foundations for

authors' research trajectory and prototyping of a gamified platform for university/college students.

2.2 Investigation

With students being exposed to more and more digital platforms where their data could be at risk, cybersecurity has become critical to higher education. Conventional cybersecurity training methods like lecturing or text-based modules cause loss of student's attention and are not engaging most, especially with non-IT students (Pramod et al., 2024). In the past few years, gamification has been acknowledged as a potential intervention to boost engagement, motivation, and retention through embedding game-like features into learning settings. There have been several studies that showed the effectiveness of learning games in terms of daily participation which fosters long-term behavioural modification by forcing students to apply their knowledge in simulated or interactive setting (Razack & Saad, 2024; Maluda et al., 2023).

The next subsections go deeper into the potential of gamification in cybersecurity training, from its definition and use for awareness raising, to an analysis of the mechanisms at work, case studies at university level, available tools and challenges around. These discussions will lay a grounding for the adaptation of gamification to address issues pertaining to university students, and validate the creation of a customised platform which is the CyberSafe Learning Hub.

2.2.1 Gamification in Cybersecurity Awareness

Gamification means the inclusion of game design elements in nongaming environments to enhance user experience and motivation. Within cybersecurity education, researchers have found gamification to enhance knowledge retention and promote security proactive behaviours (Pramod et al., 2024). Common features are challenges, levels, badges and storytelling in that they enable learners to work interactively with content as opposed to passively consuming learning. Indeed, a few institutions have started utilizing gamified modules with good student feedback in terms of enjoying and not fearing learning (Razack & Saad, 2024).



Figure 2.1: Example of gamified awareness module interface (Razack & Saad, 2024)

Figure 2.1 illustrates a sample interface of a gamified cybersecurity awareness module. The interface is a mix between visual narratives and interactive elements, such as quizzes, scoring structures and badges. The following is an overview of how the module combines these components to show how game play can enhance learning. The structure motivates learners to advance tiers, collect badges, and see a sense of accomplishment which reinforces motivation and long-term engagement.

This study has validated gamification as the primary engagement technique for CyberSafe Learning Hub. Previous research (Huotari & Hamari, 2019; Pramod et al., 2024)

shows that the addition of challenges, badges and storytelling enhances user motivation and learning. This is in line with a goal of the project to render cybersecurity learning more interactive and attractive than traditional means.

2.2.2 Mechanism of Gamified Learning

The gamification of cybersecurity is a three-part process involving content delivery, interactive engagement, and feedback. Content delivery works much like the same field, in which learners are presented with micro-lessons or scenarios that unfold. Hands-on practice can be used as collaborative quiz or simulation with role-playing. Feedback instruments, like scores or development profiles are useful for pupils to compare their growth and incentivize constant participation (Gwenhure et al., 2024). These methodologies were designed so that

learners stay engaged and encouraged, while being reminded of effective cybersecurity habits.

The second stage is interactive participation, in which learners directly use their knowledge. These can come in the form of quizzes, simulated phishing attempts, riddles or role-playing games where students play defenders against cyber-attacks. Engagement is also enhanced when the learners are placed at the centre of action, involved rather than entertained.

The third stage is feedback, and reinforcement, which is important in maintaining motivation. And leaderboards, scorecards, achievement badges and progress dashboards give users instant feedback on their efficacy. Such feedback serves to strengthen correct behaviour and identify which areas are still in need of improvement, so that learners can continue working on the material. It has been shown that if feedback is instantaneous and transparent, students will tend to persevere and acquire a sense of accomplishment (Maluda et al., 2023).

These stages, in turn, form a self-reinforcing learning cycle that boosts awareness, builds retention and drives motivation. The mechanism explains why so many institutions of higher learning are now looking at gamification: It takes cybersecurity training from being a compliance-based chore to being an engaging, even fun experience that is in tune with the students' desire for interactive and goal-oriented learning.

Seeing as gamified learning incorporates content delivery, engagement and feedback it is important to structure how learning is designed within the CyberSafe Learning Hub. Insights from Gwenhure et al. (2024) and Maluda et al. (2023) support the addition of attributes like quizzes, phishing exercises or real-time progress dashboards to the platform. These elements

help to ensure that learners remain engaged and sustainably improve their cybersafety knowledge in the CyberSafe Learning Hub environment.

2.2.3 Case Studies of Gamification in Higher Education

Recent researches indicate the importance of gamification in higher education. Maluda et al. (2023) tested gaming and cybersecurity training for undergraduates, and discovered that when the sessions were gamified this significantly enhanced understanding and memory retention when compared against not gamified lectures. Similarly, Ambikapathy et al. (2024) Games and higher education in Malaysia Gamification is found to have an impact on the increased attendance rate and motivation of young generation learning at Malaysian universities based on their learning preference. A typical element of gamification platforms are leaderboards, which have been proved in previous works to induce competition and motivation between students (Maluda et al., 2023; Ambikapathy et al., 2024). While these studies are not

directly related with the use of leaderboard, they emphasize the fact that competitive components sustain engagement and support long-term behaviour change.

Review of the case series described in Maluda et al. (2023) and Ambikapathy et al. (2024) prove that gamification can work on the educational context. These findings justify the inclusion of features such as leaderboards and progress tracking in the CyberSafe Learning Hub, which have been shown to increase engagement and motivation for university students.

2.2.4 Existing Tools and Platforms

There are already several sources that use gamification in cyber security education. For example, TryHackMe has the labs/games approach with points/levels/badges, but the content isn't beginner friendly and requires a lot of technical literacy. HackTheBox Academy offers an online environment based on virtual machines where students could solve penetration testing challenges gamified but it might be too difficult for university or college students that do not have a technical background. Cybrary, for example, offers organized cybersecurity courses that come with quizzes, hands-on labs and career-focused road maps. Cybrary is a bit more user-friendly compared to HackTheBox and predominantly emphasizes workforce training/professional certification rather than student-oriented awareness.

These constraints underscore a requirement for a dedicated platform that combines gamification with accessible, awareness-themed content targeted at university and college students.

FYP4124 Page 26 of 263

MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB



Figure 2.2: TryHackMe interface with progress badges

Figure 2.4 illustrates the learning roadmap dashboard from Cybrary, which provides structured course pathways, skill-tracking, and progress monitoring features. Cybrary, in contrast to TryHackMe and HackTheBox Academy, is more focused on career development, certifications and workforce training. Its roadmap guides learners in mapping out their path as per career requirements, so it is definitely a helpful resource for IT learners. But even though Cybrary's easy-on-the-eye layout most certainly makes accessing the content easier, it may not be the ideal solution for university entrants that are at beginner level, and need more awareness raising, gamification-based training.

The analysis of existing gamified cybersecurity platforms like TryHackMe, HackTheBox Academy and Cybrary has shown that the current tools are aimed at advanced technical audience rather than beginners. Our research supports the focus of CyberSafe Learning Hub as a starter site and awareness-centered experience specifically for tertiary education students. The results validate the course's choice to focus on low-level cyber security knowledge rather than more advanced penetration testing challenges.

2.2.5 Challenges and Limitations of Gamification in Cybersecurity Awareness

While gamification has been shown to have a great potential for promoting cyber security awareness, it also presents serious threats. A first-order limitation lies in the risk of over-focusing on extrinsic rewards such as points and badges which might make students concentrate on earning rewards rather than internalizing secure habits. This could detract from real learning to shallow competition.

FYP4124 Page 28 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

Another major challenge is the prior knowledge of learners. Some works emphasize that participants usually have very different expertise when they start a gamified training. For instance, if the task is too easy, experienced learners can get relatively bored by feeling that the task is meaningless.

If on the other hand, the activities are too challenging, students may become discouraged and eventually tune out. In both scenarios, the efficacy of gamified learning is degraded. This highlights the importance of precise balancing of difficulty, skill scaffolding, and adaptive learning paths in order to keep all learners engaged and benefiting from the training.

Additionally, contextual adaptation remains a limitation. Strategies that prove successful in corporate environments, such as competitive leaderboards, may not translate effectively into higher education settings where collaboration and inclusivity are more important (Gwenhure et al., 2024). Poorly designed gamified systems can unintentionally exclude learners or discourage participation if they do not align with the educational culture or learners'

motivations.

These challenges underline the importance of designing gamified systems that are simple, inclusive, and tailored to the learners' expertise and environment. A balanced approach that integrates both intrinsic motivation (curiosity, mastery, problem-solving) and extrinsic rewards can maximize effectiveness and foster long-term behavioural change in cybersecurity awareness.

Investigating challenges such as overreliance on extrinsic rewards, differences in learner skill levels, and contextual adaptation issues (Gwenhure et al., 2024) provides valuable guidance for improving the design of the CyberSafe Learning Hub. These insights justify incorporating adaptive difficulty levels, inclusive design principles, and a balance between intrinsic and extrinsic motivation. By addressing these limitations, the CyberSafe Learning Hub aims to create a more engaging, fair, and educational gamified learning experience for all users.

Table

Table 2.1: Investigation Summary Table

Year	Title / Author	Aim	Findings
2023	Evaluating the Effectiveness of Gamification to Increase Cybersecurity Awareness among Students — Maluda et al.	To evaluate the impact of gamification on cybersecurity awareness among students.	Gamification significantly improved awareness and engagement, showing higher retention compared to traditional teaching methods.
2024	Enhancing Cybersecurity Awareness through Gamification: Design an Interactive Cybersecurity Learning Platform for Multimedia University Students — Razack & Saad	To design an interactive gamified platform for cybersecurity education.	Students reported better motivation and learning outcomes, with increased interest in cybersecurity practices.

2024	Gamification of Cybersecurity Awareness for Non-IT Professionals: A Systematic Literature Review — Gwenthure & Rahayu	To review existing studies on gamification for non-IT professionals.	Gamification is effective in improving awareness but challenges remain in sustaining engagement and measuring behavioral change.
2021	Assessing the Effects of Gamification on Enhancing Information Security Awareness — Wu et al.	To assess the impact of gamified approaches on information security awareness.	Gamified training increased awareness and improved compliance behavior compared to non-gamified training.
2022	Game-based Learning Platform to Enhance Cybersecurity Education — Khan	To introduce a game-based learning platform for cybersecurity education.	Students showed higher knowledge retention and found the learning experience more enjoyable.
2025	A Case Study in Gamification for a Cybersecurity Education Program: A Game for Cryptography — Huitema & Wong	To develop and test a gamified cryptography learning tool.	The game improved students' understanding of cryptography concepts and fostered active participation.
2024	Gamification for Cybersecurity Education in Malaysian Universities: A Case Study — Ambikapathy et al.	To explore the effectiveness of gamification in local higher education	Reported increased participation and motivation among students, highlighting cultural and contextual relevance.
2024	Gamification in Cybersecurity Education: A State-of-the-Art Review and Research Agenda – Pramod et al.	To examine existing gamification approaches in cybersecurity education and outline future research needs.	The review highlights key game elements used, common benefits, existing challenges, and areas for further development in cybersecurity education.

2.3 Related Works

Cybersecurity training platforms like TryHackMe, Cybrary, and HackTheBox Academy have gained immense popularity for teaching the concepts of cybersecurity using gamification as well as simulations and structured learning. As much as these platforms are successful, the learning assumes that the user has technician-professional level of understanding, which does not cater well to the general university and college student population. The next subsections of this chapter discuss the platforms' methodologies, the interfaces, the strengths, and the limits of each, including relevant figures.

2.3.1 TryHackMe

TryHackMe is a gamified cybersecurity training platform designed for both beginners and advanced learners. Its content is organized into “rooms” and “learning paths” consisting of reading with guided materials, interactive activities, and lab work. Users of the platform get

motivated with elements of gamification, such as points, badges, and leaderboards.



Figure 2.5: TryHackMe Dashboard

As illustrated in Figure 2.5, the TryHackMe dashboard is where users can track their progress and other metrics such as earned badges, streaks, and leaderboard ranks. This demonstrates the ability of the platform to provide elements of gamification intended to provide a sense of achievement and competition.

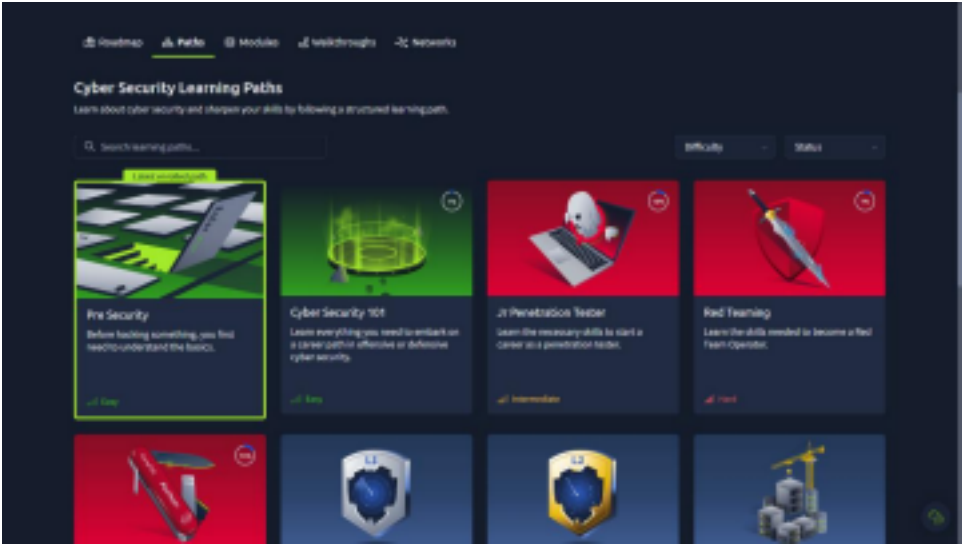


Figure 2.6: TryHackMe Learning Path Page

Figure 2.6 illustrates the Learning Path page that shows a list of rooms based on levels or categories. This illustrates how the system organizes progression of knowledge. It teaches students step-by-step, from surface to deep knowledge.

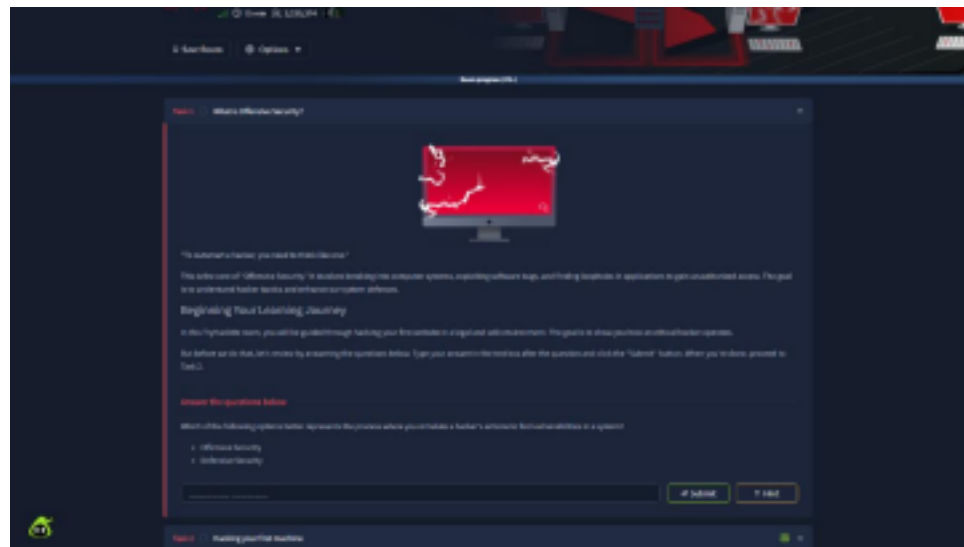


Figure 2.7: TryHackMe Challenge Room Interface

An example of a Challenge Room interface is shown in Figure 2.7, where instructions are presented on one face and there may be an embedded terminal or exercise space on the other side. We think this is a great example of how both theory and practice appear to be integrated which presents the opportunity for learners apply their knowledge in real time.

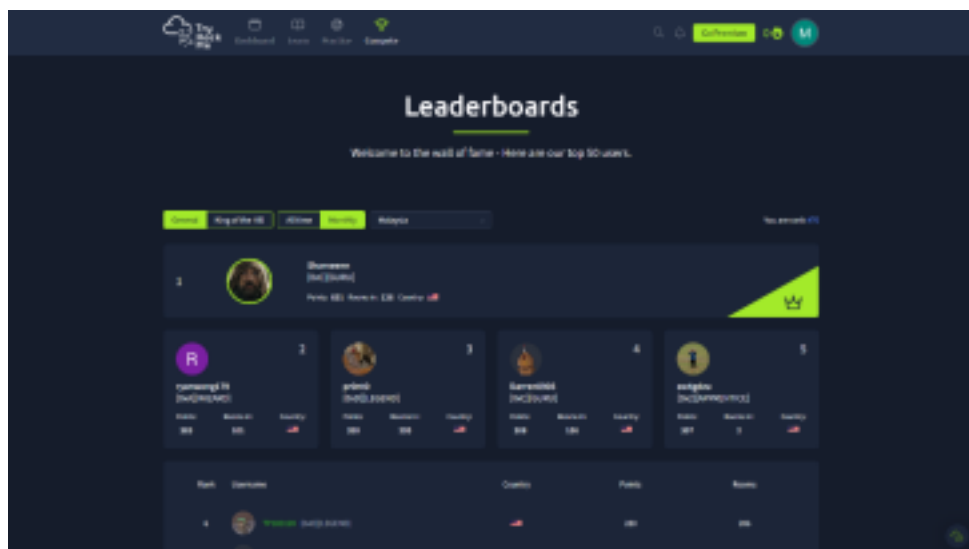


Figure 2.8: TryHackMe Leaderboard Interface

The TryHackMe's leaderboard section screenshot is shown in Figure 2.8, where users are awarded points through completing the challenges that are being displayed as toppers of the table. This competitive nature serves to increase extrinsic motivation in the learners and to keep them coming back for more.

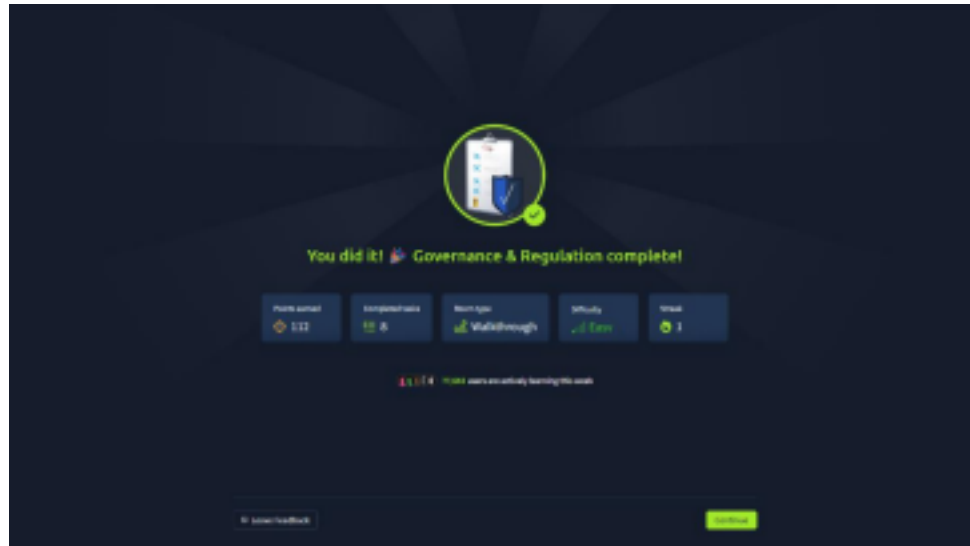


Figure 2.9: TryHackMe Completed Room Example

Figure 2.9 shows the completion summary page of a TryHackMe room known as “Governance & Regulation.” The interface shows points earned, number of tasks finished, room type and difficulty and learning streak. This solidifies the game principle of feedback and recognition that learners can see their great effort in their easy-to-read progress report.

The user starts by selecting a room or a path. Starting with a brief concept overview, rooms deliver tasks that require learners to interact with sandbox.sh environments, such as executing commands, or taking advantage of a misconfigured service. Learners submit flags or badges, as evidence of progress; they instantly earn feedback to accompany reward points. These points are aggregated into badges and ranking levels which can be viewed in the dashboard, to keep learners motivated.

TryHackMe really shines in realistic simulation with its large collection of practical rooms which mirror real-world security setups and situations. These learning labs are particularly important for practical skill development, specifically in areas such as vulnerability identification, exploit techniques and secure system configurations all of which complement theoretical knowledge through hands-on use. This is in line with prior work where the platform's gamified aspects like points, badges, ranks have also helped maintain learner motivation through providing an incentive for continued engagement and effort. But when it comes to awareness training like among non-technical university or college students, the

depth and complexity of the platform can be intimidating. A lot of the exercises take for granted knowledge of basic command-line utilities, networking material and penetration testing techniques which might exclude some less IT literate learners. On the other hand, even if technical capabilities are reinforced through offensive and defensive challenges, the platform does not directly provide simplified awareness packages like phishing recognition, password hygiene or safe web browsing which would be essential for everyday cybersecurity behaviour. In summary, TryHackMe is efficacy in up-scaling skill acquisition for budding cybersecurity professionals - albeit with a large amount of modification to the tool for it to be used as an awareness mechanism within global higher education.

2.3.2 Cybrary

Cybrary is a platform that provides structured, “course style” training in ever-expanding topics ranging from ethical hacking, to cloud security, critical incident response (cyber crisis management), network defense and threat intelligence. Is focuses on comprehensive content coverage through a combination of video, readings, quizzes and optional labs.

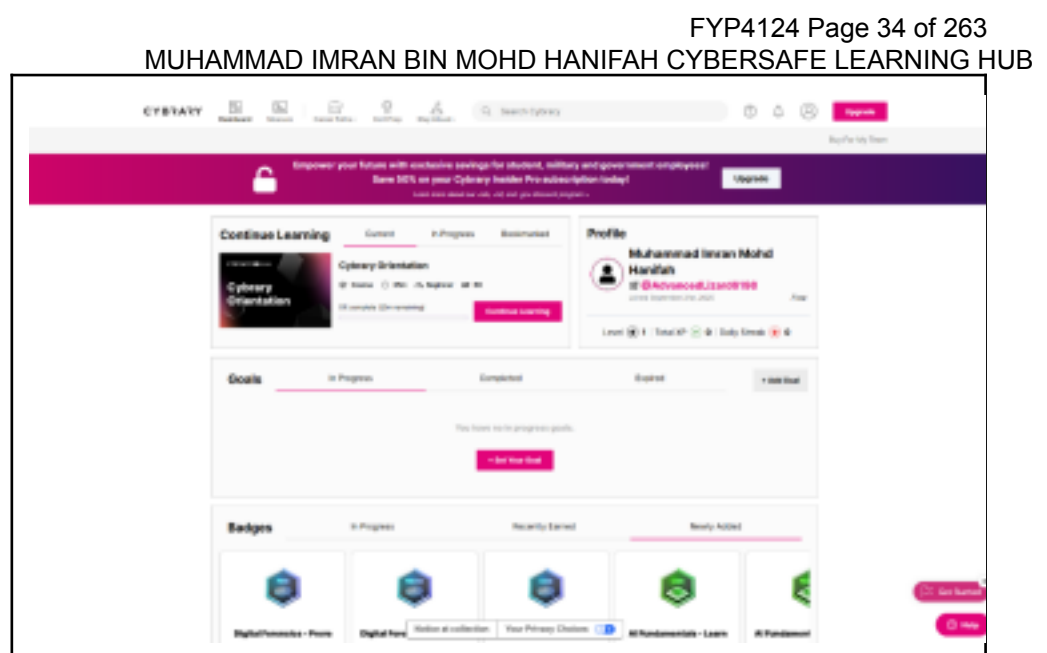


Figure 2.10: Cybrary Dashboard Page

Figure 2.10 shows the Cybrary dashboard through which users can monitor their learning progress, continue courses and develop course recommendations in line with user objectives. By visually displaying progress, achievements and next activities of learning paths

this personalised way of learning promotes a continuous involvement.

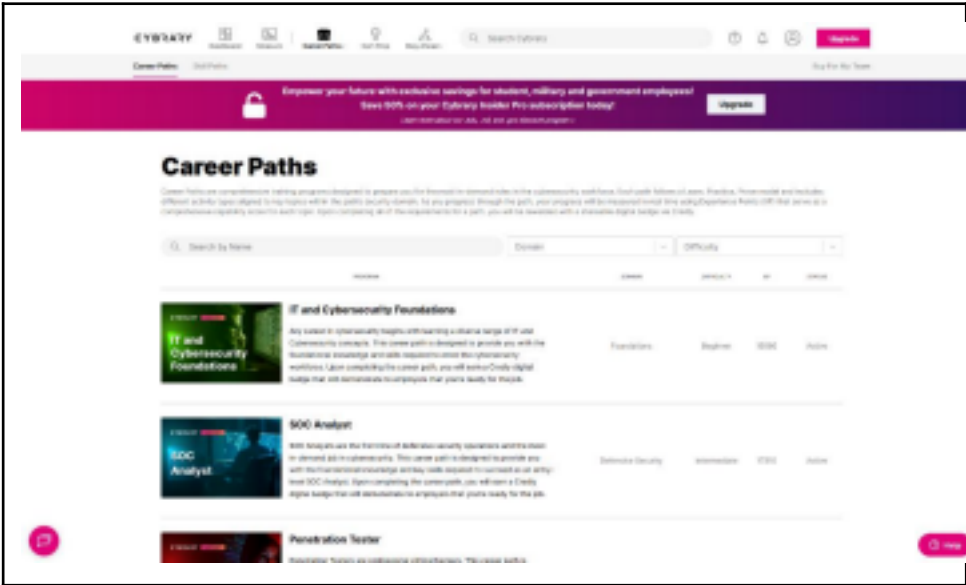


Figure 2.11: Cybrary Course Catalog and Career Paths

In Figure 2.11, a snapshot of the Cybrary course catalog and career paths is provided in which learners may follow a structured path such as SOC analyst or PenTester. This function places Cybrary among the platforms providing content that also steers its learners towards targeted professional roles in cybersecurity.

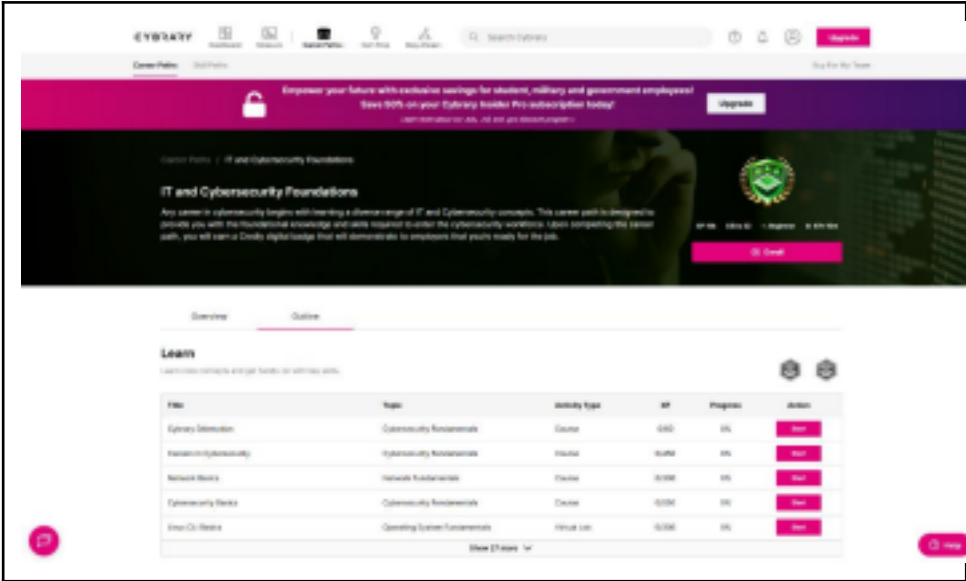
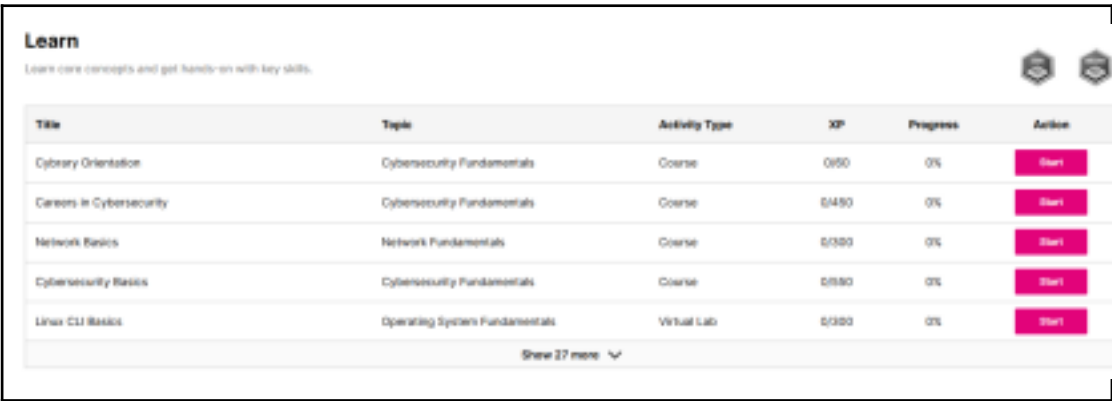


Figure 2.12: Cybrary Course Module Page

Figure 2.12 shows a Course Module page integrating video, transcripts, and checkpoint quizzes. In presenting material in various forms, and providing checks on understanding at intervals, Cybrary takes an academic approach with its structure that is close kin to traditional

classroom instruction.



The dashboard shows a list of learning activities with columns for Title, Topic, Activity Type, XP, Progress, and Action. The activities listed are Cybrary Orientation, Careers in Cybersecurity, Network Basics, Cybersecurity Basics, and Linux CLI Basics. Each activity has a 'Start' button. A 'Show 27 more' link is at the bottom.

Title	Topic	Activity Type	XP	Progress	Action
Cybrary Orientation	Cybersecurity Fundamentals	Course	0/90	0%	Start
Careers in Cybersecurity	Cybersecurity Fundamentals	Course	0/490	0%	Start
Network Basics	Network Fundamentals	Course	0/390	0%	Start
Cybersecurity Basics	Cybersecurity Fundamentals	Course	0/890	0%	Start
Linux CLI Basics	Operating Systems Fundamentals	Virtual Lab	0/390	0%	Start

[Show 27 more](#)

Figure 2.13: Cybrary Learning Progress Dashboard

Figure 2.13 shows the Learner Progress Dashboard, where data like completion percentage, and badges earned is tracked. This tracking mechanism functions to support self regulated learning by providing feedback on attainment of goals and opportunities for improvement.

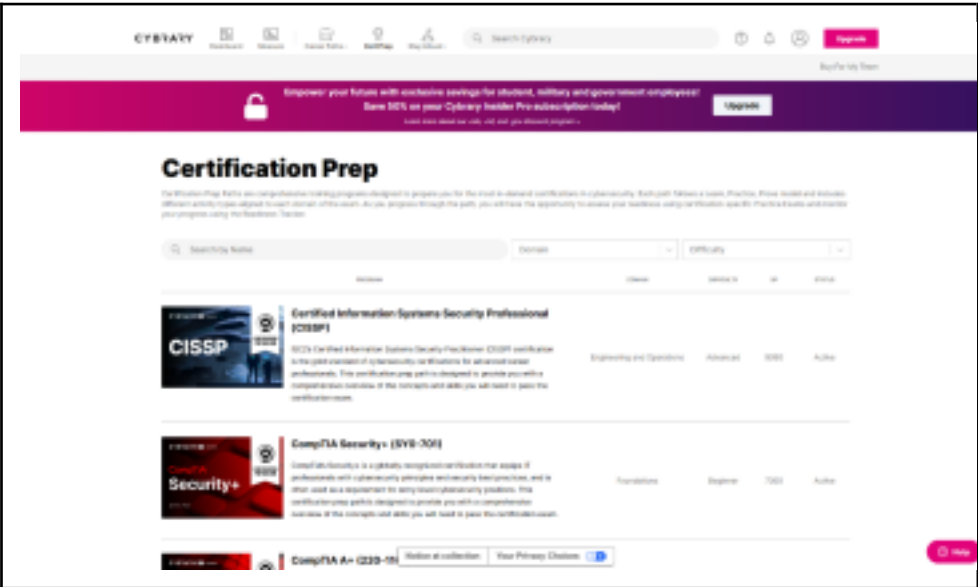


Figure 2.14: Cybrary Certification Preparation Page

Figure 2.14 is the Cybrary certification prep page, where you can find learning paths for various cybersecurity certifications including CompTIA Security+ and CISSP. This design demonstrates how Cybrary incorporates professional certification readiness into its learning

path.

Learners enrol on a course or career pathway then progress through a series of modules that integrate theoretical material with assessment. Labs are available for advanced students to test theoretical material. It is assessment driven and tracking of course progress and learner performance through quizzes, dashboards that are fairly easy to for the user to view.

Cybrary does well to itself as its courses are completely organized and detailed, offering cyber professionals to study on a clear path, knowing at all times where they stand and how much they have learnt. The platform's modular nature, checkpoint quizzes and optional labs make it to ensure that the learners gain both the theoretical knowledge and practical skills, more so for individuals aspiring professional certifications. Though, because Cybrary has a focus on courses and certification-based content, the users are not going to have as interactive simulation like TryHackMe or HackTheBox. This can limit the hands-on involvement and can therefore be less suitable for non-technical or a typical college population who are looking for shorter, engaging awareness training. Therefore, though Cybrary is perfect for those already in the workforce as far of career learners are concerned, in practice its methodology is also not enough for people that learn better with scenario-based learning by doing activities on security awareness.

2.3.3 HackTheBox Academy

HackTheBox Academy brings the HackTheBox ecosystem into structured learning paths focusing on penetration testing labs with a curated list of resources. It is a favourite among high-level learners and cybersecurity experts.

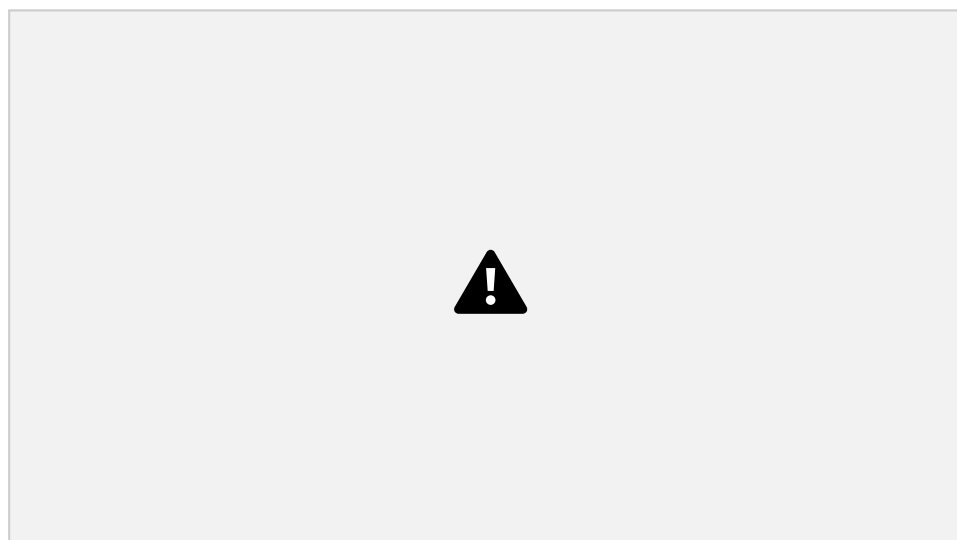


Figure 2.15: HackTheBox Academy Dashboard Page

Figure 2.15 displays the HackTheBox Academy Dashboard, where users can view accumulated points, ranks, and achievements. This gamified interface encourages motivation through competition and visible progress tracking.

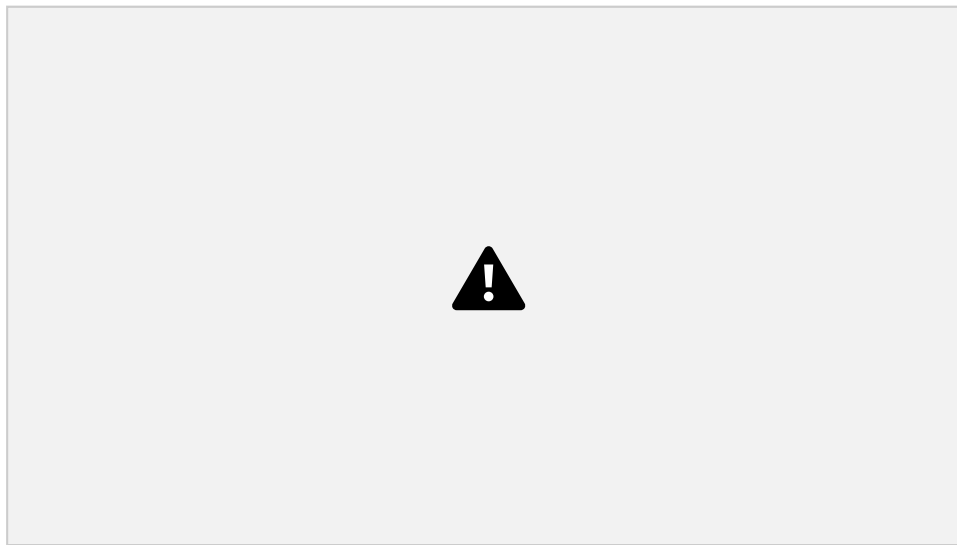


Figure 2.16: HackTheBox Academy Lab Interface Page

Figure 2.16 shows the Lab Interface direction, where learners interact with a set of vulnerable virtual machines or services for exploitation. This design makes the most sense

FYP4124 Page 38 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB
and is based on real penetration testing environments, which will enable students to be transferred from this space to a real world with ease.

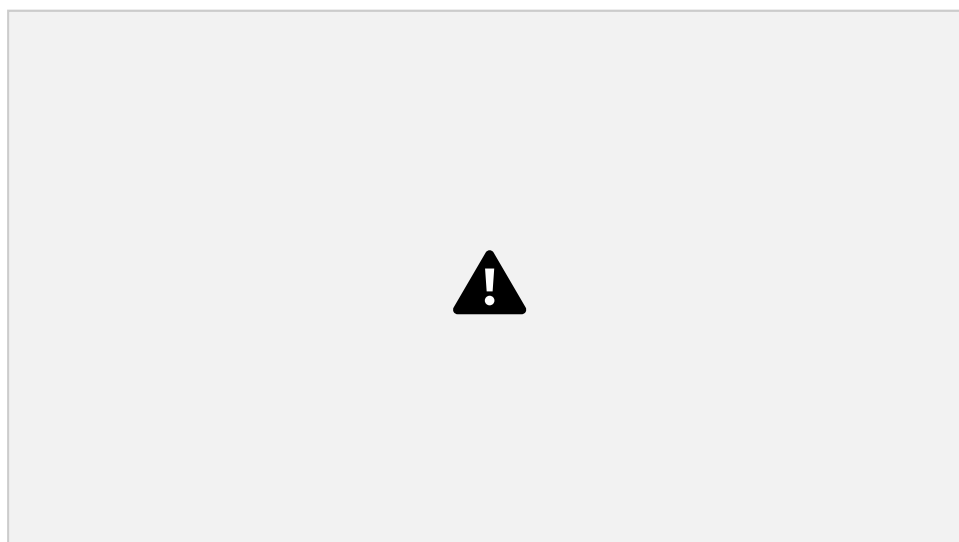


Figure 2.17: HackTheBox Academy Completion Badge Page

Figure 2.17 shows the Learning Track Completion Page which gives learners digital badges when they have completed certain modules or tracks. This badges are used in reward and acknowledge the achievement and skill development.

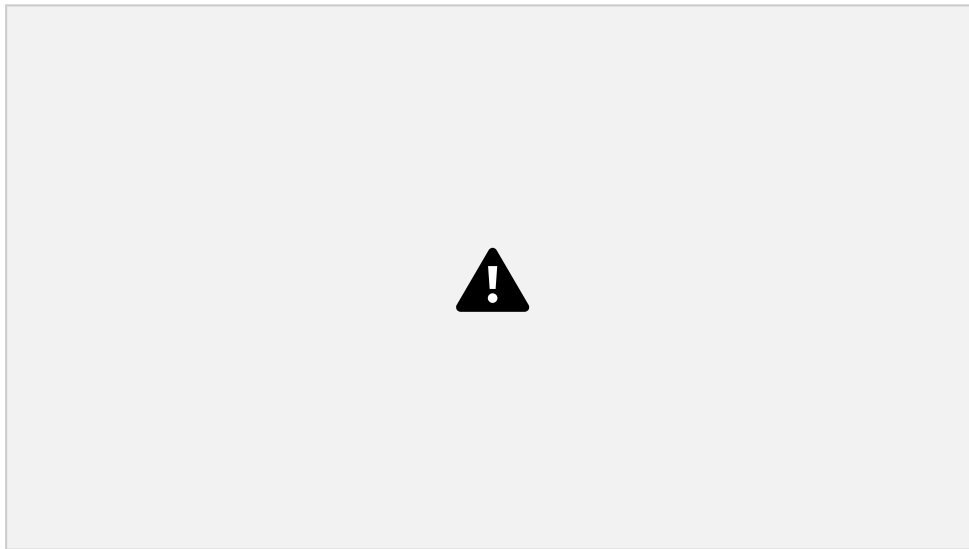


Figure 2.18: HackTheBox Academy Certificate Page

Figure 2.20 shows the Certificate Page, which learners can use to see or save certificates of completion for particular learning paths or modules. This formal recognition provides greater satisfaction and credibility to learners.

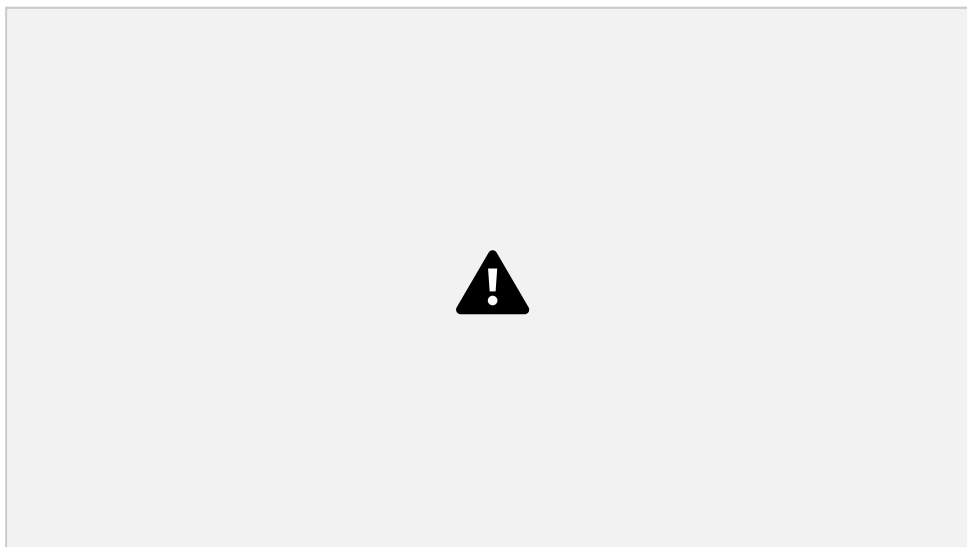


Figure 2.19: HackTheBox Academy Exams Page

Figure 2.21 shows the Exams Page to enable learners take guided assessments that access their understanding and empirical competence. Examination encourages retention of skill and application in real world.

HackTheBox Academy is modeled with a lab-first method. Students are provided with a quick high-level briefing before they attempt to solve a vulnerability challenge in a sandboxed setting. Advancement is based on finding a “flag,” indicating that the task has been executed. Learners progress in rank by solving increasingly challenging labs, and have the option to

challenge friends in team competitions.

HackTheBox Academy is truly a one of its kind when it comes to being real life like labs having very close replica of actual vulnerabilities and real-world attack vectors. It's a scenario based learning approach where participants will encounter a variety of practical security challenges ranging from web vulnerabilities to network attacks, in an environment that closely simulates your computing work stations and your networks. This practice-based approach is an efficient method of building technical skills, problem solving applications and draw insights into adversary techniques. The platform's high technical intensity and reliance on previous knowledge does not seem suitable for mass university or college student awareness training. It offers a depth that's very helpful for advanced students, but may be too high of a ceiling for non-technical students on the labs. However, HackTheBox Academy is a good model for example for scenario-based training that might be modified or even simplified to make the cybersecurity awareness engaging for wider groups of people.

MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB **2.4 Comparison**

It is also worth comparing online cybersecurity learning platforms in respect of features, target users, usability, emphasis on security and learning outcomes. There are also other training platforms such as TryHackMe, Cybrary and then HackTheBox which all offer their own unique path toward cyber security training from guided labs, cert prep, and full-on advanced hacking simulation. A comparison between these 3 platforms is in the table below with key criteria are taken into consideration.

Table 2.2: Comparison of Existing Project

Criteria / Project	TryHackMe	Cybrary	HackTheBox
Features	Hands-on labs, guided learning paths, gamified challenges, beginner to advanced content.	Video-based courses, structured career paths, certification prep, broader IT training.	Capture-the-flag (CTF) style labs, pro labs, real-world hacking simulations, community-driven challenges.
Users	Students, cybersecurity beginners, and professionals seeking practical labs.	IT and cybersecurity learners, professionals preparing for certifications, corporate teams.	Intermediate to advanced cybersecurity enthusiasts, penetration testers, red teamers.
Navigation	User-friendly dashboard, task-based modules, gamified progress	Traditional e-learning platform, course catalogs, structured paths.	Technical interface, challenge boards, VPN-based access to labs.

	tracking.		
Security Element	Focus on hands-on security skills, attack/defense scenarios, real tools.	Theory + conceptual security knowledge, compliance, certification focus.	Realistic hacking environments, exploit development, offensive security practice.
Output	Practical experience, strong foundation for beginners, progression into advanced labs.	Certification readiness, theoretical understanding, corporate-level upskilling.	Advanced hacking expertise, strong offensive skills, recognition in global CTF community.
Accessibility	Beginner-friendly, low prior knowledge required, web based interface.	Accessible via standard web browser, moderate tech skills required.	Requires intermediate technical knowledge, VPN setup, and familiarity with Linux/command line.
Gamification	Points, badges, leaderboards, streaks to motivate learning.	Minimal gamification, mostly progress tracking and completion metrics.	Leaderboards, points, ranking in global CTF challenges, competitive environment.
Adaptability	Flexible learning paths, self-paced, suitable for classroom integration.	Fixed courses, career tracks, less flexible for self-directed exploration.	Highly scenario-specific, advanced labs may be adapted for competitions or skill practice but less beginner-friendly.

2.5 Discussion

By comparing TryHackMe, Cybrary and HackTheBox, it is clear that every platform demonstrates specific strengths which cater for different aspects of cybersecurity education.

TryHackMe has structured learning paths which are accessible and guided. This makes it particularly effective for beginners or intermediate learners. Through its interactive labs and game-like design, engagement is maintained as well as foundational skills gradually increase in level. These elements will be essential for classroom use or even formality within universities under the leadership of researchers working on thematic projects. The features that will be implemented from TryHackMe are simple gamified elements such as points, badges, and basic guided learning paths to keep learners motivated and on track.

In contrast, Cybrary is more about professional development and preparing yourself for industry standard certifications. With its wide range of courses covering both the technical and managerial sides to security practices in today's businesses world, it is perfect for corporate training environments as well as certification-seekers themselves seeking industry-recognised creds. The features that will be implemented

from Cybrary are progress tracking dashboards and short quizzes to monitor learning outcomes effectively.

HackTheBox Academy however is a completely different experience, focusing on realistic penetration testing and extreme problem-solving. Within this platform, it contains the natural appeal of Capture-the-Flag style activities and a grassroots ecosystem built by the community for learners who have already obtained fundamentals and want to further develop their skills in offensive security. The HackTheBox Academy features that will be added to CyberSafe Learning Hub are some basic, scenario based exercises and a simple leaderboard to facilitate interaction and have some friendly competition.

In general, choice of a platform is determined by the learner's objectives. TryHackMe shows that it is most useful for intermediate or advanced learning, Cybrary does best in certification or career focused pathing, and HackTheBox Academy is more beneficial to someone looking to do advanced attacks and keep honing the "offensive" skill set. Between them, these three platforms combine to cater to the entire range of cybersecurity learning from basic awareness all the way through professional development and specialized skills.

2.6 Conclusion

This chapter was a survey of prior works for the use of gamification in cybersecurity awareness, and other learning platforms with interactivity, gamified practice or scenario-based learning. Research shows that gamification leads to improved engagement, motivation and retention of learners but there are issues such as pre learning requirements and contextual adaptation which needs to be accounted for. TryHackMe, Cybrary, and HackTheBox Academy are some of the platforms analyzed to identify some divergences. TryHackMe have guided and gamified learning paths for beginners. Cybrary focuses on professional growth and certification. HackTheBox Academy serves excellent realistic challenges for advanced learners.

The features that we will be embracing for the CyberSafe Learning Hub aims primarily in simplicity and low barriers to use, consisting of guided learning paths, points and badges to support gamification, tracking of progress, short quizzes as well as basic scenario-based exercises. It is expected, through this combination, that the project will be able to deliver to university and college level students a user-friendly useful system conveying a range of cyber security issues as well as being practical and easy to use.

3 METHODOLOGY

3.1 Introduction

Methodology is the general strategy that outlines systematical design, development and evaluation of a project. It provides a roadmap that helps the authors organize their work to meet specific goals efficiently. An effective methodology defines how software development stays organized, traceable and adaptive over time (Pressman & Maxim, 2020).

Agile Methodology was used in developing the CyberSafe Learning Hub. Agile has a focus on adaptability, teamwork and incremental development which is more appropriate for web-based systems that are developed with tight user feedback loops. As students and lecturers are both to be involved actively in the project, Agile will provide for quick responsiveness when it comes to their feedback and at the same time a structured as well as purposeful development process.

3.2 Agile Methodology

Agile methodology is a software development practice that promotes continuous incremental progress in an iterative fashion (sprints). There are several stages connected to the sprint, which is planning, design, development, testing, deployment review and launch cycle that iterates throughout the project to provide flexibility and continuous improvement.

According to Popa et al. (2021), the Agile methodology offers flexibility and communication between developers and clients by splitting the project into small portions that can be improved in each iteration. This allows for review of the solution along with feedback, which reinforces continuous work to improve quality and match what users want. Agile tends to be transparent, promotes communication and gives teams the opportunity to produce working editions of the system quickly, mitigating risk with software projects.

This approach is particularly appropriate for this project as there is regular input from both the students (end users) and lecturers (administrators). With short iteration cycles feedback can rapidly be collected and reflected upon which not only keeps the platform engaging, interactive and educational but also effective. Furthermore, Agile allows flexibility in refining gamification features, lesson modules, and progress tracking elements based on real user experiences.

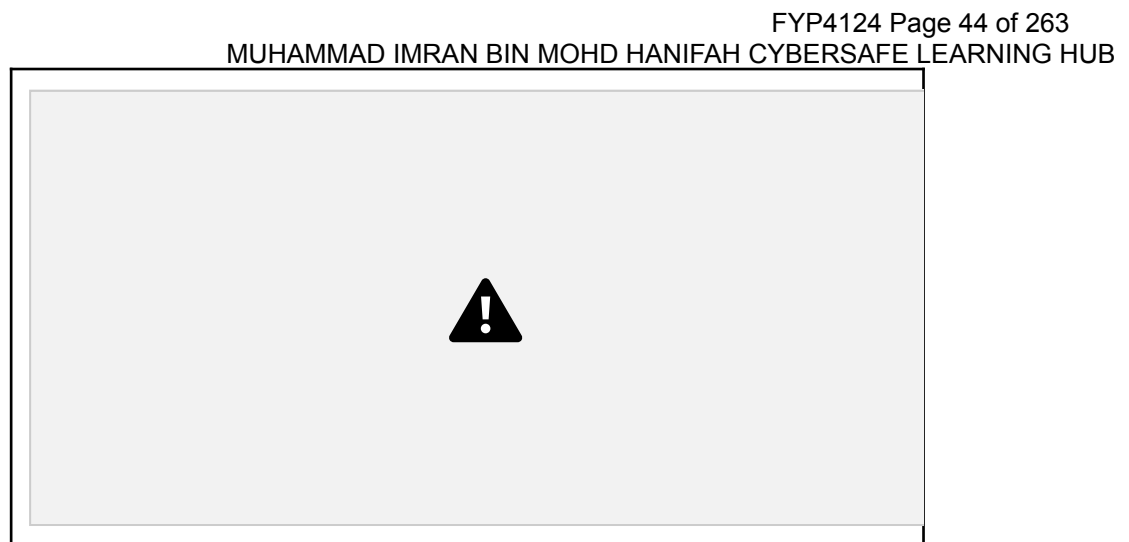


Figure 3.1: Agile Methodology Diagram (Popa et al., 2021)

The process for the Agile methodology can be seen in Figure 3.1, illustrating how development moves through stages of a number of sprints. Every sprint has a generic Plan -> Design -> Develop -> Test -> Deploy -> Review -> Launch that is repeated, leaving plenty of space for iterations and small porting periods. This process provides the space for each phase to reinforce system utility and user satisfaction, which is in line with the project's focus on developing an agile and user-centred cybersecurity learning platform.

3.3 Phases in Agile Methodology

The agile methodology is a process which is represented by discrete and iterative phases that enable to control the software development in an efficient way. The CyberSafe Learning Hub will participate in five essential phases for the project: Planning, Design, Development, Testing and Deployment and Review. There would be deliverables for each stage that would work in unison with the system to make it successful.

3.3.1 Phase 1: Planning

The planning stage is the cornerstone of Agile development, and it concentrates on being inspired by user and client-based needs before any designing or implementing starts. The first stage of this research examined the gap regarding cybersecurity awareness between college students and how they preferred to learn. The primary focus was to ensure that the design of CyberSafe Learning Hub would be driven by user requirements and education objectives.

Interview was conducted with a lecturer of Kolej Teknologi Antarabangsa Cybernetics (KTAC) who also acts as the client and somebody who is familiarize in teaching regarding cyber security to retrieve accurate data which are related to the project. The interview drew attention to the problems in teaching awareness of cybersecurity and what the lecturer expects from interactive learning platform. This conversation led to identifying indispensable system properties like tracking progress, gamified curriculum and role-based access control.

FYP4124 Page 45 of 263

MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

In addition, a survey was distributed to students asking them about their current knowledge of cybersecurity features like online threats and interest in gamified learning. The students' feedback has oriented and shaped what will be incorporated into the system, in terms of its functional requirements and contents or activities included according to their learning styles.

According to the discoveries obtained from these data collection activities, appropriate technology was selected for system development such as Laravel 12 for backend development, MySQL for database management, Bootstrap for front-end design and Google OAuth to ensure secure authentication. The output from this phase is the signed off proposal document, a comprehensive system requirement statement and technology selection report that form the starting point for Agile phases to follow.

3.3.2 Phase 2: Design

In this step architecture and user interface are defined. Flowcharts, use case diagrams and database schema were created to ensure that the development proceeds with systematic approach. Wireframes for students and lecturer dashboards were also created in order to visually explain the concept of lessons, quizzes and gamification. This stage is needed to make sure the back-end and front-end designs fit with user needs and are in line with project objectives.

3.3.3 Phase 3: Development

The development cycle is characterized by continuous coding and integration. The primary features like authentication, role-based dashboards, lessons and challenges CRUD operations, and gamification based on points/badges and progress tracking are implemented in Laravel. Continuous testing happens after each sprint so that all the modules are working fine and bug free. Its outcome is a prototype of the CyberSafe Learning Hub.

3.3.4 Phase 4: Testing

The testing phase is to guarantee the system reliability, security and usability. Several checks are conducted such as unit, integration and system testing. PHPUnit will be used to ensure that the tested module functions as expected with the notifications and data flow. A test with lecturers and students is used to test usability and provide feedback for refinement.

3.3.5 Phase 5: Deployment and Review

After testing, the system is launched in a testing or "staging" environment for trial by real users. Feedback from users and lecturers is gathered with surveys and interviews. Based on this

FYP4124 Page 46 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

feedback, the system will be improved for performance and user experience. When approved, the final version will be deployed.

3.4 Requirement

The requirement phase is about collecting both functional and non-functional requirements so that the system satisfies users and client needs. According to Awati (2023), the requirement phase is vital for a project's development as it provides an understanding of requirements for contextual needs, conditions and expectations that must be addressed if the project is to succeed. This procedure aims at determining what the CyberSafe Learning Hub is expected to do and/or how the tooling should behave in order to fulfil these expectations in an efficient manner.

If the requirements are well-captured, then we can develop and test the system in a methodical way with little chance for misunderstanding or technical error. The need for this project was determined through direct feedback, collected via questionnaires and interviews to ensure the final product aligns with user needs and learning goals.

3.4.1 Data Gathering Techniques

Questionnaire and interview have been used to determine the needs of the users and their acceptability. These methods produce quantitative and qualitative results, encompassing an extensive exploration of user requirements and preferences in cybersecurity learning.

3.4.1.1 Questionnaire

Questionnaire is a structured tool developed for collecting opinions, experiences and knowledge from individuals in large numbers (Bhandari, 2021). For the current study, a questionnaire was administered to at least 50 students in Kolej Teknologi Antarabangsa Cybernetics (KTAC) using Google Form.

The questionnaire comprised of questions related to students' current knowledge on cybersecurity typical for a user, their preferred method of studying and learning process and the interest over gamified learning.

3.4.1.2 Interview

An interview is a qualitative research tool that helps to discover a participant's perspective through direct communication and discussion (University of Bath, 2023). An interview was conducted for this project with a lecturer of the KTAC acting as the client's side.

The interview was aimed at an exploration of the lecturers' issues with teaching cybersecurity subjects and some thoughts on how the interactive learning method may make better student participation. The lecturer stressed that classical teaching elements, like already available slides or lectures are not very attractive when it comes to attraction, and offered the recommendation of introducing interactive modules for learning experience such as quizzes and gamification.

3.4.2 Functional Requirement

Functional requirements describe the tasks and or activities CyberSafe Learning Hub should be able to do. These requirements describe what the system must do to achieve the user and client requirements in an efficient manner.

Table 3.1: Functional Requirement Table

Functional Requirement	Description
User Authentication	Allows users to register manually or log in using Google OAuth.
Role-Based Access Control	Provides different access rights for students and lecturers.
Content Management	Allows administrators to create, edit, and delete categories, modules, lessons, quizzes, and quiz questions.
Access to Lessons and Quizzes	Enables students to view and complete cybersecurity awareness modules.

Quiz Scoring and Feedback	Automatically evaluates quiz submissions, enforces an 85% passing threshold, and displays detailed results with correct and incorrect answers.
AI Chatbot Assistance	Provides an AI-powered chatbot during quizzes to help students understand cybersecurity concepts without revealing direct answers.
Progress Tracking	Records and displays each student's learning progress.
Gamification Features	Awards points and badges for completing lessons and quizzes.
Dashboard Monitoring	Enables lecturers to monitor and review student performance.

3.4.3 Non-Functional Requirement

Non-functional requirements specify the quality properties of the system, and include performance, security, usability, reliability and scalability.

Table 3.2: Non-Functional Requirements Table

Category	Requirement
Performance	The system should load each page within 5 seconds under normal conditions.
Usability	The interface must be simple and user-friendly for non-IT students.
Reliability	The system should maintain consistent operation without data loss.
Security	User credentials and progress data must be encrypted and protected.

3.4.4 System Requirement

System requirements delineate what hardware and software are required for the CyberSafe Learning Hub to develop and run. These requirements are there to make sure a system works effectively, meets its functional and non-functional objectives and is easy for users.

Since this project is related to designing a web-based system which could include interactive and gamified characteristics, so it would be mainly recommended to use the reliable hardware and software which should able to possess capability of using tools for web development, local server operations and maintaining database on proper note.

3.4.4.1 Hardware Requirement

Hardware requirements are the physical components that support system implementation and operation. Good hardware guarantees smooth processes from coding to testing and deploying,

without any hiccups or bottlenecks.

In this project, the Lenovo IdeaPad Gaming 3 laptop are used for development, which is powerful enough to perform web programming, executing virtual environments and testing databases.

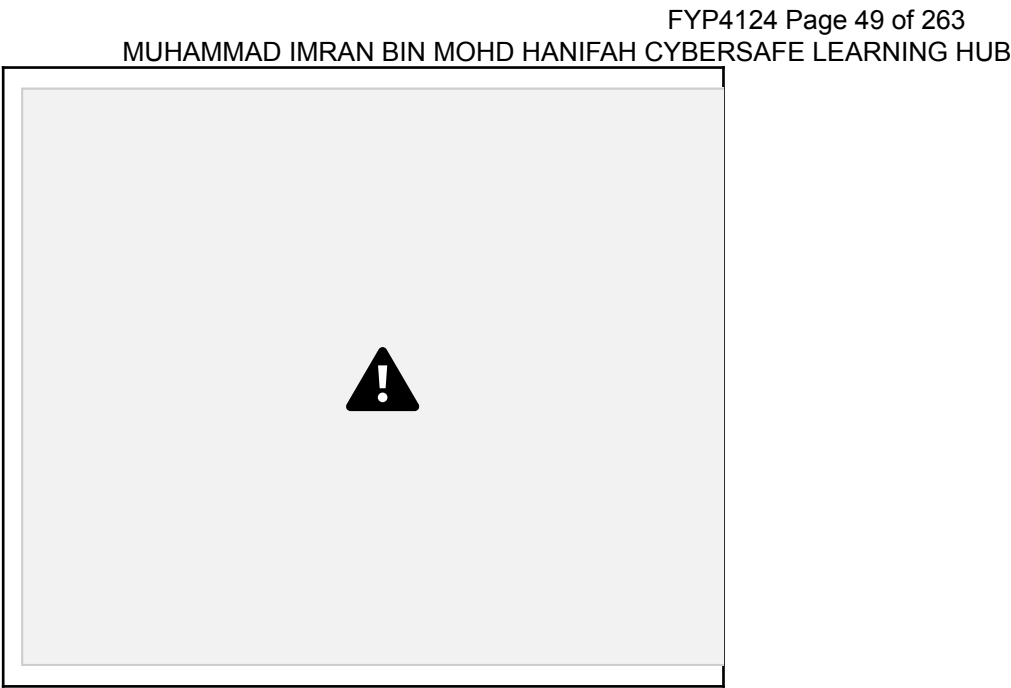


Figure 3.2: Lenovo IdeaPad Gaming 3

Figure 3.2 shows the Lenovo IdeaPad Gaming 3 which is the laptop that will be used to develop this project.

Table 3.3: Hardware Requirements Table

Specification	Description
Model	Lenovo IdeaPad Gaming 3
Processor	AMD Ryzen 5 5600H (6 Cores, 12 Threads, up to 4.2 GHz)
Graphics	NVIDIA GeForce GTX 1650
RAM	16 GB DDR4
Storage	512 GB SSD

Operating System	Windows 11 (64-bit)
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The Lenovo IdeaPad Gaming 3 is a mix of performance and efficiency that developers can work with. The AMD Ryzen 5 5600H CPU provides fantastic multitasking, and fired up with local testing and code compilation is a breeze thanks to 16 GB of DDR4 RAM. The built-in NVIDIA GeForce GTX 1650 GPU can deliver web visuals and modern browsers efficiently. Additionally, having a 512 GB SSD speeds up files access and project loading times, while Windows 11 ensures compatibility with current web development environments such as XAMPP, Visual Studio Code, and Laravel.

FYP4124 Page 50 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

3.4.4.2 Software Requirement

The software requirement is the technical tool or framework that needs to be used to construct, test and operate on CyberSafe Learning Hub. Every software component has a role to play in backend development, frontend design, database management and authentication process.

- **Code Editor: Visual Studio Code (VS Code)**

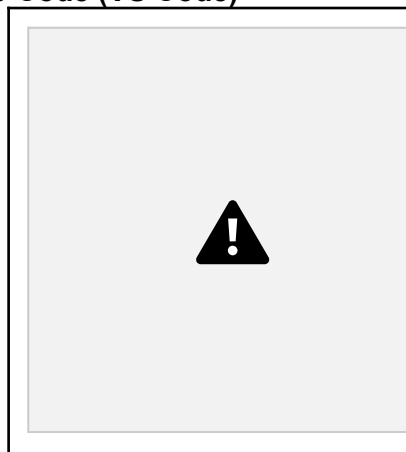


Figure 3.3: Visual Studio Code Logo

The primary source-code editor, was Visual Studio Code. It is compatible with languages like PHP, HTML, CSS and JavaScript and comes with built-in debugging, syntax highlighting, version control etc. These features better streamline workflow and promote enhanced code quality in Laravel development.

- **Backend Framework: Laravel 12**

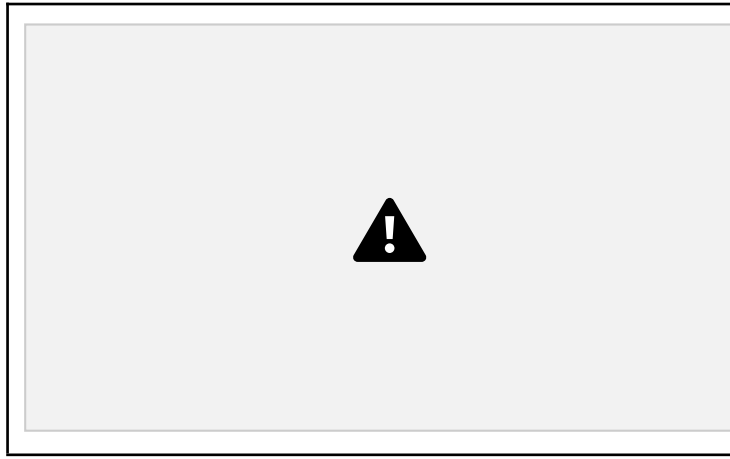


Figure 3.4: Laravel 12 Logo

The backend framework utilized is Laravel 12 which provides a secure and robust way to authenticate, perform database operations as well as MVC architecture.

Its functionalities, middleware, routing and validation reduce the complexity of creating secure web applications.

- **Database Management System: MySQL 8.0**

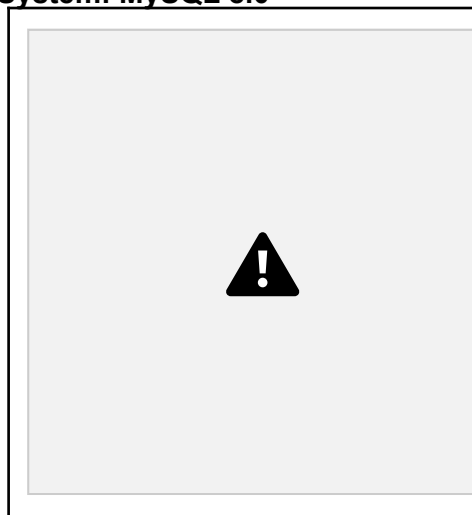


Figure 3.5: MySQL Logo

The relational database management system (RDBMS) we use is MySQL 8.0 to save all user data, lesson content, quiz results and gamification records. It supports many features including reliability, scalability, and it's compatible with Laravel's Eloquent ORM.

- **Frontend Framework: Bootstrap 5**

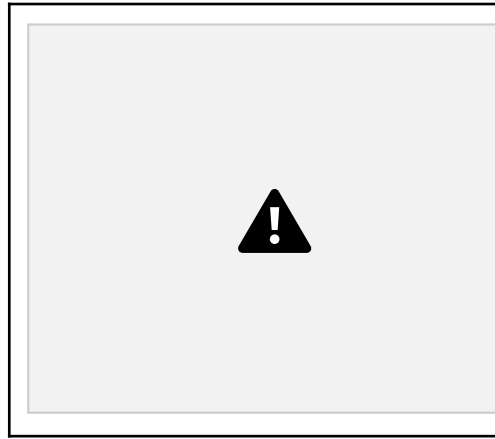


Figure 3.6: Bootstrap Logo

Bootstrap 5 will be used for responsive and user-friendly layout design. This is to ensure that pages are easy to navigate and visually appealing to the users.

- **Authentication API: Google OAuth**

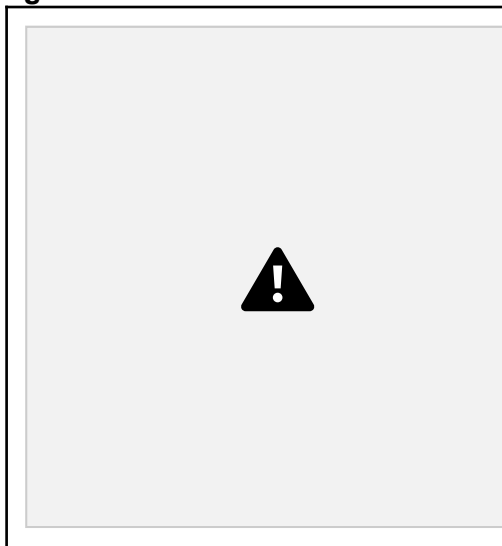


Figure 3.7: Google OAuth Logo

Google OAuth API enables secure authentication by using a third-party identity provider and allowing them to sign in with their Google account. This makes documentation easy to access, and lessens the risk of password-based vulnerabilities.

- **Testing Framework: PHPUnit**

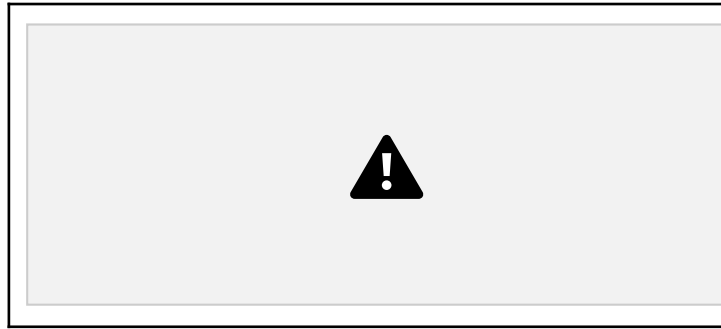


Figure 3.8: PHPUnit Logo

PHPUnit will be used for unit and integration testing to complete the application with efficient outcome. Automations built through testing not only help in ensuring reliability but also in early development cycle's issue.

• **Web Server Environment: XAMPP (Apache)**

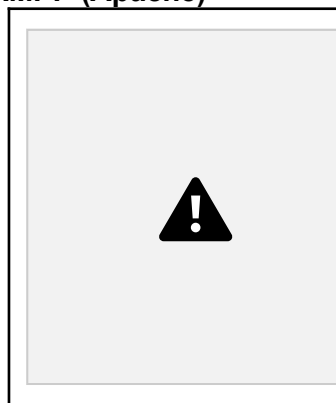


Figure 3.9: XAMPP Logo

XAMPP is a relatively easy to use local web server environment containing Apache, MySQL and PHP. It is designed to enable running the Laravel application locally before pushing to the cloud.

3.5 Analysis

The interpretation of the data collected from our requirements gathering can be determined in the analysis phase as it is medically analysed and produced, delivering sound and stunning insight for

developing the CyberSafe Learning Hub system design. Based on questionnaires given to students and a interview with a lecturer from Kolej Teknologi Antarabangsa Cybernetics (KTAC), this study attempted to learn about users level of awareness of the cyber security implications around them, their learning preference and what kind of system abilities they required in an online education system. A questionnaire generated quantitative data that showed the students are aware of what cybersecurity information is, yet lacks a lot of practical knowledge and has inconsistent safe habits when online to keep them safe. In the meantime, the interview provided some qualitative points from an educator's perspective, including a desire for structured and interactive content for lessons but also tools to track students' progress through features like quizzes and game elements designed to boost engagement.

3.5.1 Data Gathering Analysis

Data collecting and analysis is the process of scrutinizing hard information gathering method and drawing a conclusion based on meaningful insights which system development will use (Saunders et al 2023). This analysis was based on questionnaires and structured interviews, collected for the CyberSafe Learning Hub project. The primary goal was to learn about the requirements, awareness and expectation of target users which is students and lecturers from Kolej Teknologi Antarabangsa Cybernetics (KTAC).

FYP4124 Page 54 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

Using this analysis, the study pinpointed crucial system prospection for critical assets of interest such as cybersecurity awareness material, interactive learning modules and engaging user functionalities. Through a systematic review and synthesis of the collected data, we identified the features, functionalities, and design framework required for creating a user-centric and functional platform. These captured learnings are crucial learning points that will contribute to shaping the CyberSafe Learning Hub, enabling it to meet user needs and add value for students as they navigate cyber security and digital safety.

3.5.1.1 Questionnaire Analysis

A questionnaire was used as a quantitative data collection method through online platform using Google Forms to KTAC students. The survey instrument had 20 close-ended questions on a range of issues organized into four thematic categories that encompassed demographics, awareness about cybersecurity, preferences of learning style and system requirement. The survey was released over the period of 13th—27th October 2025 and resulted in a total of 73 valid responses.

The questionnaire served two goals. First, it sought to confirm that the problem exists through an assessment of students' cybersecurity awareness and behaviours. Findings indicated that students generally had weak cybersecurity awareness and habits, especially concerning password management and phishing prevention activities. This supported our belief in a learning solution

such as CyberSafe Learning Hub. Second, this questionnaire aimed to identify system and design requirements such as students' preferred learning styles, motivational factors related to gamification, and essential non-functional aspects such as usability, security, and system performance.

Quantitative results from this pre-development phase more broadly informed specific design requirements for the CyberSafe Learning Hub that were relevant with identified gaps in awareness, and fulfils user's needs.

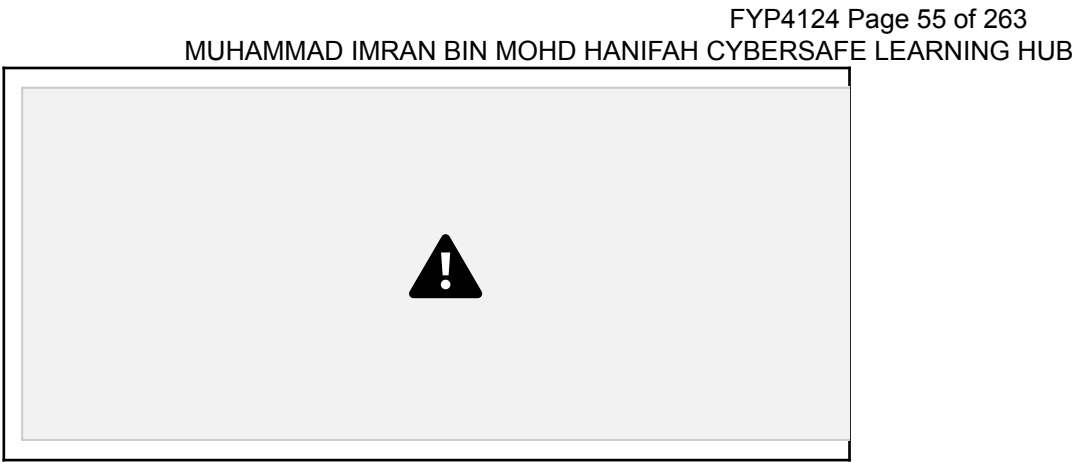


Figure 3.10: Age of Respondents

The age distribution of respondents is displayed in Figure 3.10. Majority are 24–26 years old (34.2%) followed by 21–23 years old (31.5%), 18–20 years old (20.5%) and >26 years of age (13.7%). This suggests that most respondents fall into the average college student age group, which is reasonable to explore the cybersecurity knowledge for college students.

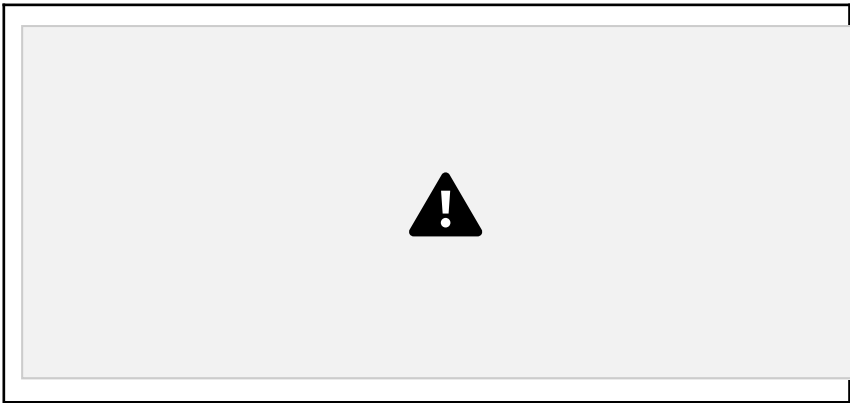


Figure 3.11: Gender of Respondents

Figure 3.11 shows the relative proportion of male and female respondents, with a higher level of males at 53.4% than females at 46.6%. The almost equal gender split ensures that both the male and female reactions are fairly represented, so the results aren't slanted towards one or the other.



Figure 3.12: Program of Study

The respondents' program of study is illustrated in Figure 3.12 with 49.3% of the participants are studying IT-related, while the remaining 50.7% from another than IT-related field. It is based on a well-balanced proportion that the awareness of cybersecurity is important in all fields, which has inspired the project to engage IT and non-IT students as the target audience for such a platform.

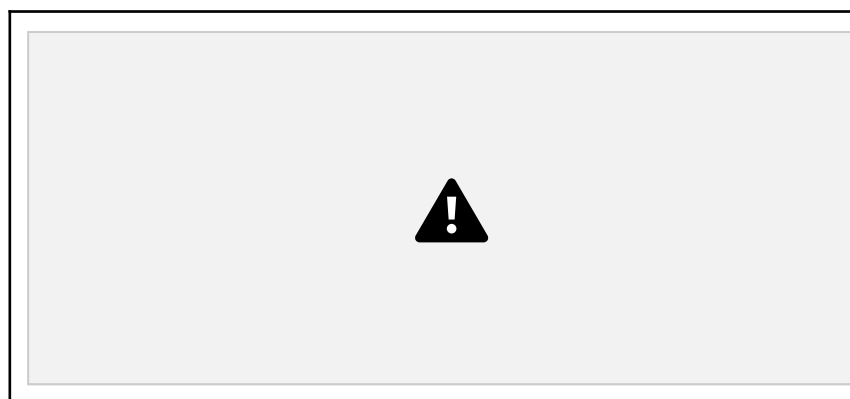


Figure 3.13: Frequency of Internet Use

Figure 3.13 shows the extent to which students used Internet. In terms of daily internet usage, 35.6% reported using the computer for 2-4 hours, another 35.6% used it for 5-8 hours, and 15.1% used it for less than two hours and further use by only 13.7%, who spent over eight hours per day on the computer. The results of these data suggest that internet users who participated in the study are highly at risk to be exposed to cyberthreats and the concept of awareness ceaselessly enforce their importance.

FYP4124 Page 57 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB

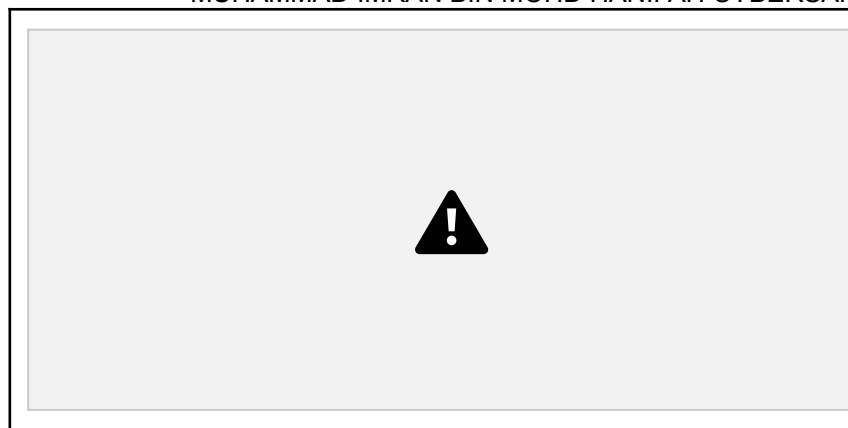


Figure 3.14: Level of Cybersecurity Knowledge

Figure 3.14 illustrates the current level of awareness amongst respondents in basic cybersecurity, like strong passwords and phishing. The findings show that 45.2% rated their knowledge as very low, 32.9% low, 16.4% high, and 5.5% met the 'very high' threshold. It reveals that majority of students are less knowledgeable on cybersecurity and underscores the need for educational interventions in order to make them well informed about online safety.

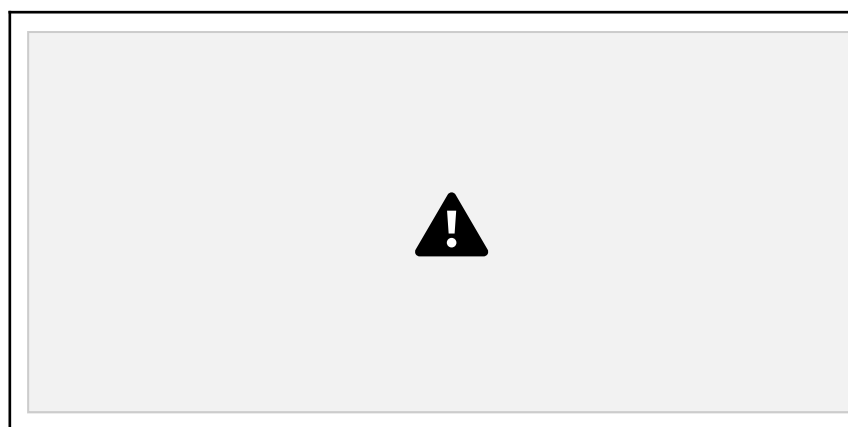


Figure 3.15: Experience of Cybersecurity Incidents

Figure 3.15 shows whether an individual has fallen victim to a cyber-attack such as phishing, malware, or hack. The overall prevalence of such incidents was 76.7% and 23.3% did not experienced them. This suggest that students are consistently under threat of cyber security which highlights the importance of developing awareness and safe online behaviour practices.

FYP4124 Page 58 of 263
MUHAMMAD IMRAN BIN MOHD HANIFAH CYBERSAFE LEARNING HUB



Figure 3.16: Participation in Cybersecurity Awareness Programs

Figure 3.16 displays the frequency of respondents attending cybersecurity awareness programs/training. Results show as 43.8% attend such programs occasionally, 31.5% frequently, 17.8% rarely, and 6.8% never. This implies that some students have been exposed to cyber security learning but the majority of them do not regularly engage in awareness programs.

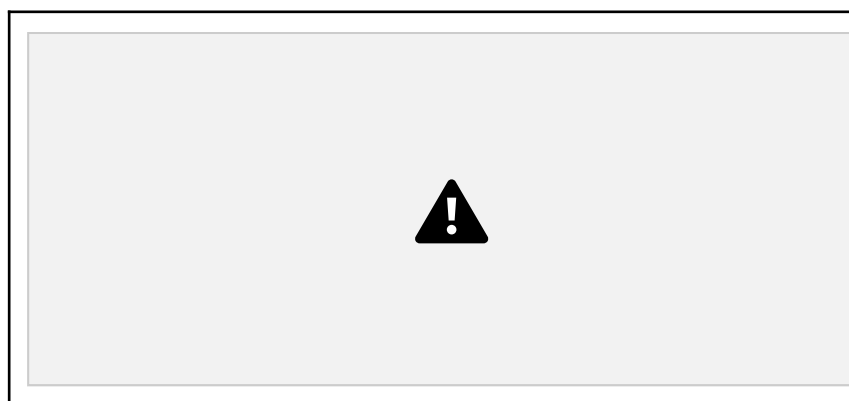


Figure 3.17: Importance of Cybersecurity Awareness

Figure 3.17 displays the answers of the survey as whether cybersecurity awareness is important for university students or not. Most, 86.3%, replied Yes and the rest, 13.7%, said No. This substantial concordance indicates that students understand the importance of cyber-security awareness as an integral part of their digital education.



Figure 3.18: Confidence in Protecting Personal Data

Figure 3.18 shows respondents' confidence in the security of their personal data and online accounts. The numbers are 45.2% confident, 35.6% very confident, while 11% of respondents are not confident, and another 8.2% are only slightly confident. These findings suggest that the vast majority of college students exhibit confidence but may have an overestimation of their ability to defend themselves from threats without any formal training in cybersecurity.

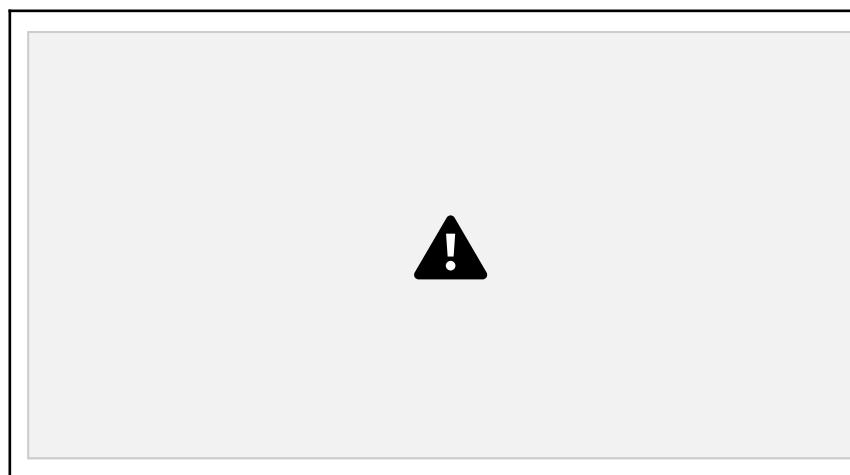


Figure 3.19: Preferred Learning Methods for Cybersecurity

The learning type preferred by the responses of what learning is there for cybersecurity are illustrated in Figure 3.19. Quizzes were the most preferred teaching tool (83.6%), and this was followed by videos (64.4%), interactive modules (56.2%), articles (53.4%) and gamified learning (51%). This indicates that learners prefer interactive and participatory learning, which justifies the integration of gamified and quiz content into the CyberSafe Learning Hub.

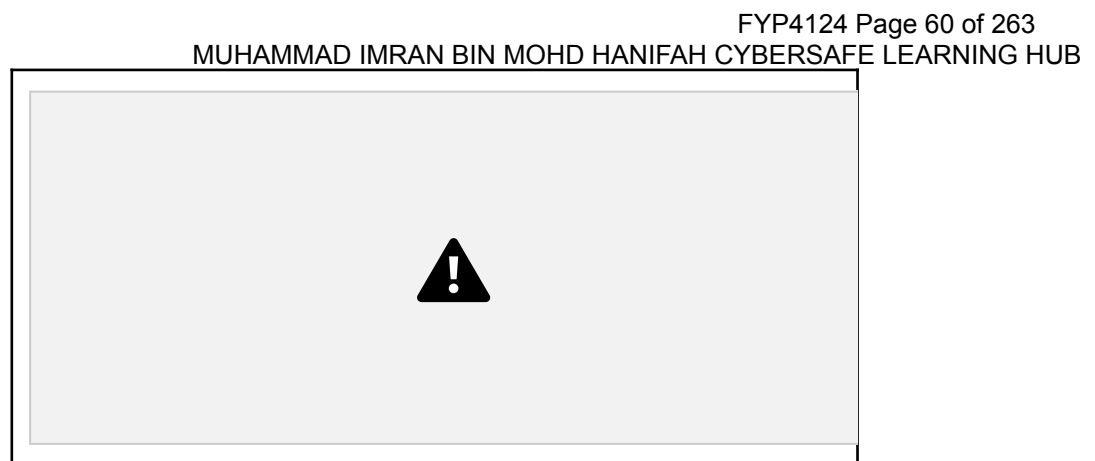


Figure 3.20: Frequency of Applying Safe Online Practices

Figure 3.20 shows how often respondents apply safe online practices. The results indicate that 37% stated they always follow safe practices, 35.6% said often, 11% said sometimes, 12.3% said rarely, and 4.1% never apply them. This suggests that while a majority of students consistently maintain online safety, a small group still neglects basic security habits, indicating a need for continuous reinforcement of safe online behaviors.

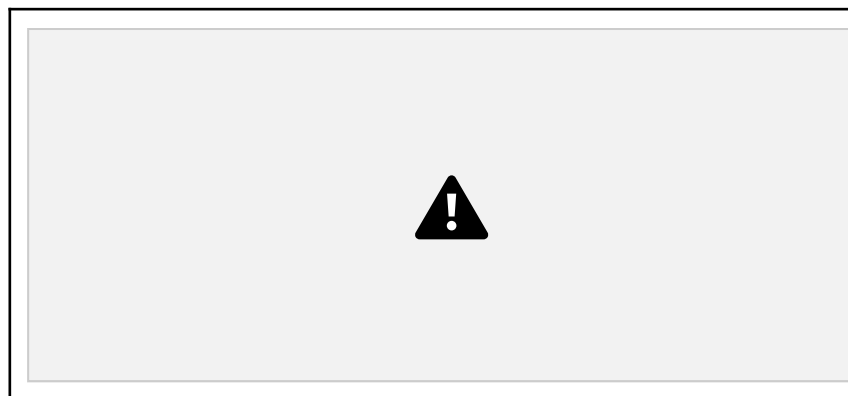


Figure 3.21: Awareness of Security Practices

Figure 3.21 displays respondents' familiarity with cybersecurity activities such as using two

factor authentication or regularly updating their passwords. It was responded Yes by 76.7% and No by 23.3%. Thus, this shows that the majority of students know of these practices but almost one fourth still do not know about them.

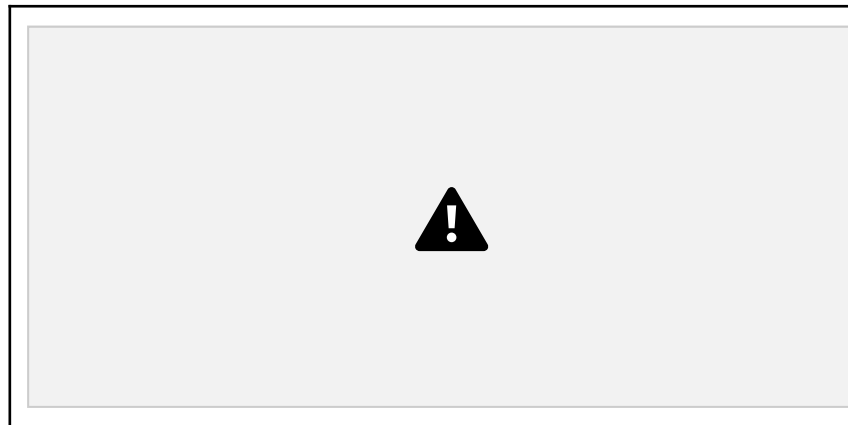


Figure 3.22: Interest in Using Cybersecurity Learning Platforms

Figure 3.22 demonstrates that participants are interested in using an online tool which helps providing them knowledge about cybersecurity related issues as well as education in this regard. There appear to be 76.7% that are interested and 23.3% not interested according to the data. This would indicate a great degree of acceptance and willingness on the part of students to work with a learning platform such as the CyberSafe Learning Hub.

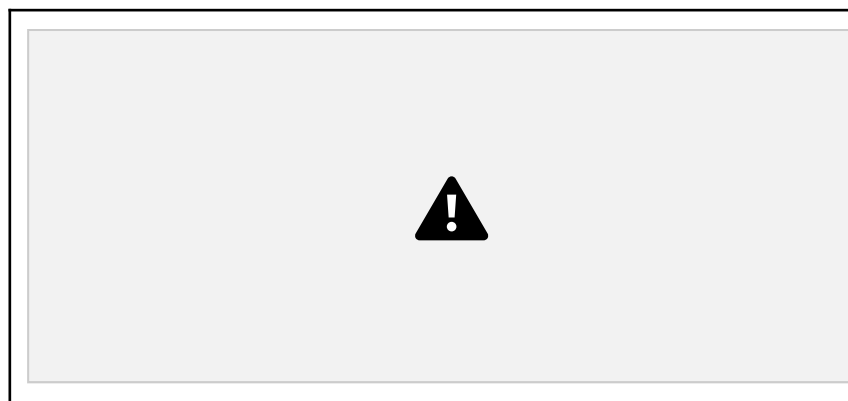


Figure 3.23: Importance of User Authentication

The significance of user authentication on a cybersecurity platform is depicted in Figure 3.23.

Results show that 38.4%, rated it very important, while 34.2% important, and other rated as moderately important (12.3%) slightly important (8.2%), not important (6.8%). That would suggest that most are aware of the value of secure access and identity validation.

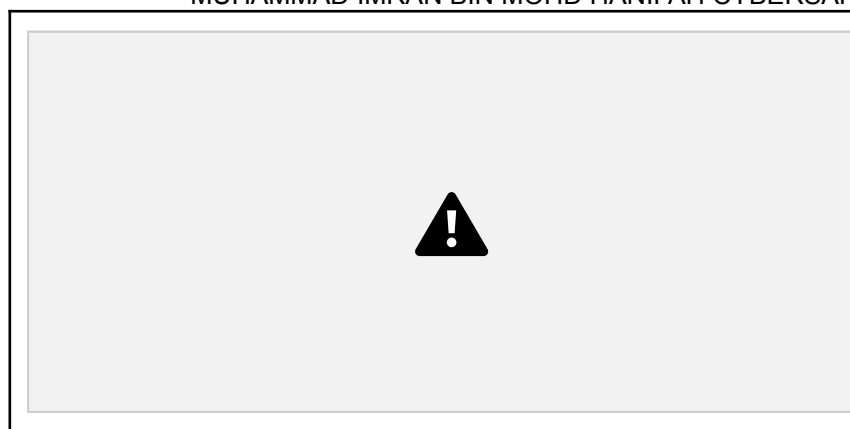


Figure 3.24: Motivation Through Gamification

Figure 3.24 shows how participants perceived gamification elements like points, badges, and challenges as motivating them. The results have shown that 37% find them very motivating, 38.4% motivating, 19.2% slightly motivating and only 5.4% not motivating. These results imply that integrating the game design elements can greatly motivate learner and make them enjoy learning about cybersecurity.

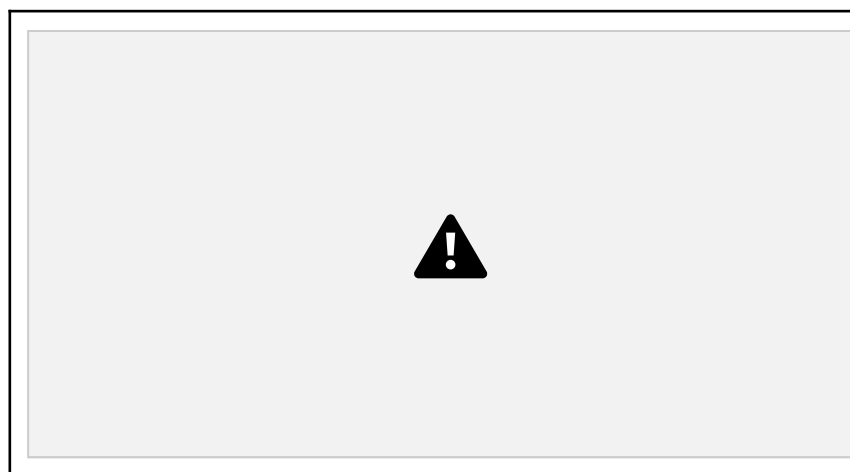


Figure 3.25: Preferred Features in the CyberSafe Learning Hub

Figure 3.25 shows what respondents consider to be the most valuable options from which they can choose for inclusion in a CyberSafe Learner Hub. The findings show that quizzes (78.1%) are the highest-demanded feature, and they are followed by leaderboards (72.6%), tutorials (58.9%), progress tracking tools (57.5%), and gamified lessons (47.9%). It means that students prefer interactive and competitive elements in their learning.

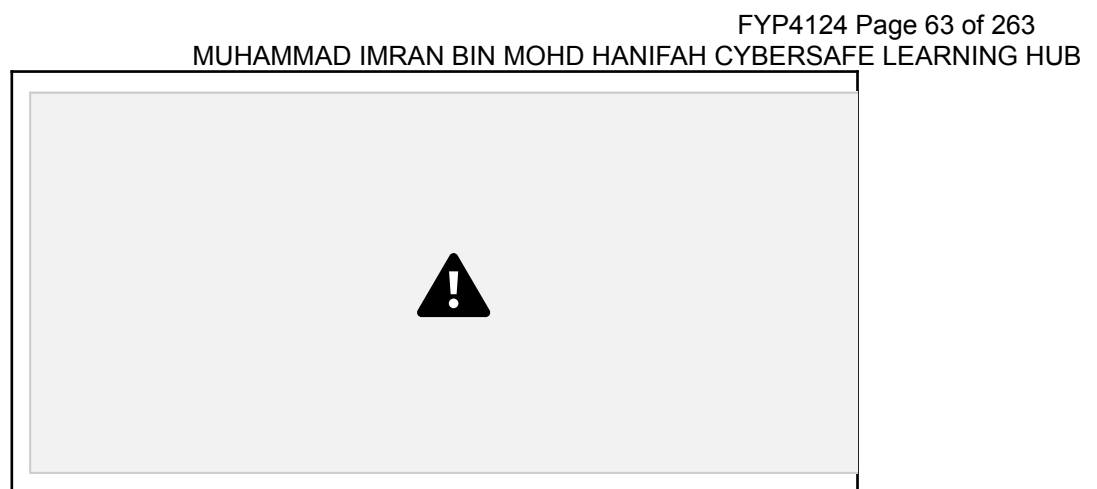


Figure 3.26: Usefulness of Progress Tracking

Figure 3.26 How useful students thought it would be to track learning progress and achievements using the platform. The results indicate that 37% rated it as very useful, 35.6% as useful, 15.1% as somewhat useful, and 9.6%, slightly useful; with only 2.7 % rating it as not useful. This indicates robust support for monitoring progress features in which learners can track improvement.

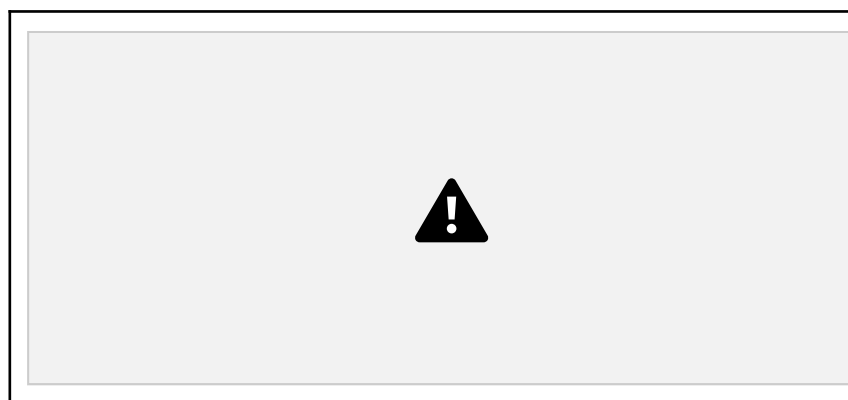


Figure 3.27: Importance of Platform Loading Speed

Figure 3.27 shows the importance of web platform loading speed according to respondents. The figures indicate that 38.4% valued it as important, 35.6% very important, 12.3% moderate

importance, 8.2% low importance and 5.5% not important to their studies respectively. This illustrates that the system performance is a basic user expectation.

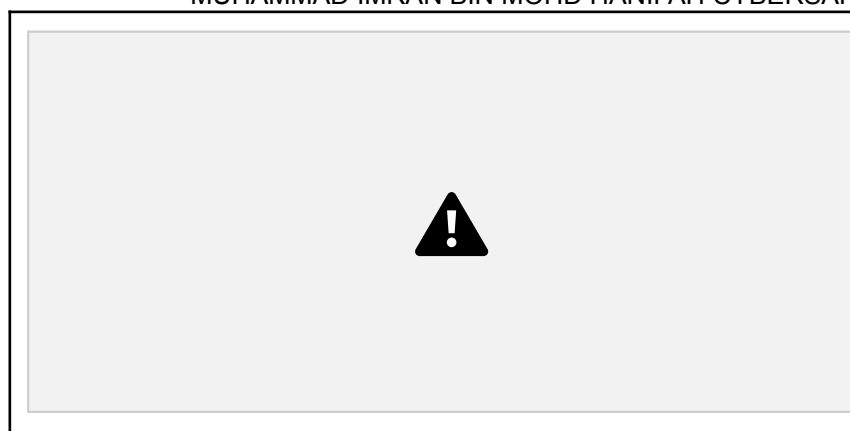


Figure 3.28: System Usability and Interface Design

Figure 3.28 exhibits the agreement among students with respect to an interface that is simple and user-friendly. Strongly agree and agree was 42.5% each, disagree was 8.2%, strongly disagree, 6.8%. This validates the fact that ease of use is important for non-IT students.

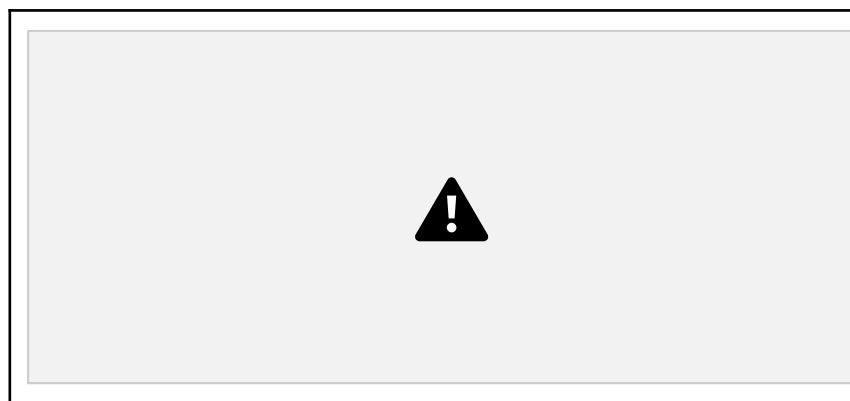


Figure 3.29: Importance of Data Security and Encryption

Figure 3.29 demonstrates the attitudes of students on data security, in example, whether learning data should be encrypted and stored securely. The agreement was 78.1% with disagreement of 21.9%. This also highlighted the need to encrypt and securely manage user data in the CyberSafe Learning Hub.

3.5.1.2 Interview Analysis

Interview with Mr. Syaheedullah Bin Yahya, lecturer of Diploma in Information Technology programme at KTAC on the 9th October 2025. He was chosen for this interview because of his professional knowledge in the information technology field and mentoring students about digital protection. This interview was aimed at collecting expert opinions on the significance of cyber security awareness on college students, and obtaining recommendations on the CyberSafe Learning Hub platform. His insights have been able to give us valuable inputs in terms of current issues we faced, effective learning approach and practical recommendation that could come into places in context of cybersecurity being taught at higher institutions.

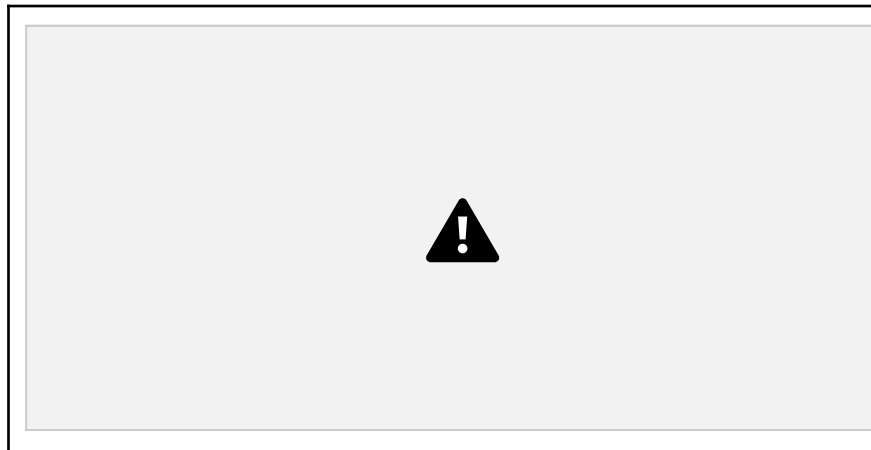


Figure 3.30: Interview session with Mr.Syaheedullah

Table 3.4: Analysis of Interview Questions

Question	Answer	Analysis
In your opinion, how important is cybersecurity awareness for college students?	• Very important in today's digital era. • Many students lack awareness of online risks.	Emphasizes the growing need for continuous cybersecurity education among students who are active internet users.
What are the most common cybersecurity challenges or issues you observe among students?	• Weak or reused passwords. • Careless information sharing. • Clicking on suspicious links/emails. • Failure to update devices.	Highlights behavioral issues as the main source of vulnerabilities, stressing the need for habit-based awareness programs.
Have you encountered any cases where students were affected due to a lack of cybersecurity knowledge?	• Yes, cases of phishing and account takeovers. • Students lost access to email or social media.	Demonstrates the real-world consequences of low cybersecurity literacy, supporting the case for more preventive education.

How effective do you think the university's current cybersecurity awareness efforts are?	• Efforts exist but limited. • Campaigns conducted once or twice per semester. • Need for continuous engagement.	Suggests that awareness efforts should shift from periodic campaigns to ongoing engagement for lasting impact.
What learning methods do you find most effective in teaching students about cybersecurity?	• Practical, hands-on activities. • Simulations and real world exercises. • Less theory, more application.	Endorses experiential learning and simulation-based training as the most effective teaching approach.

Question	Answer	Analysis
Do you think a platform such as the CyberSafe Learning Hub could improve students' understanding of cybersecurity?	• Promising idea if similar to TryHackMe. • Encourages active, self-paced learning. • Needs evaluation after implementation.	Supports gamified, interactive learning for engagement and practical understanding of cybersecurity.
What specific features or functions would you recommend to make the platform more engaging for students?	• Gamification (badges, levels, leaderboards). • Visually appealing interface. • Interactive and varied quizzes.	Recommends motivational elements and user-friendly design to improve engagement and knowledge retention.
From an administrative perspective, how important is Role-Based Access Control (RBAC) in managing lecturer and student permissions within the system?	• Very important for access management. • Ensures proper authorization based on user roles. • Prevents data misuse.	Reinforces the need for structured access control to safeguard data and maintain system integrity.
Should cybersecurity awareness be integrated across all academic disciplines, or only within IT-related courses?	• Should be across all programs. • Every student handle digital data. • Not limited to IT students.	Advocates for inclusive cybersecurity education applicable to all fields, not just technology.

What advice would you give to ensure the CyberSafe Learning Hub supports learning and long term awareness?	• Update modules regularly. • Use real-world examples. • Involve lecturers and integrate with coursework.	Suggests that platform sustainability depends on consistent updates, curriculum integration, and lecturer participation.
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3.5.2 Use Case Model

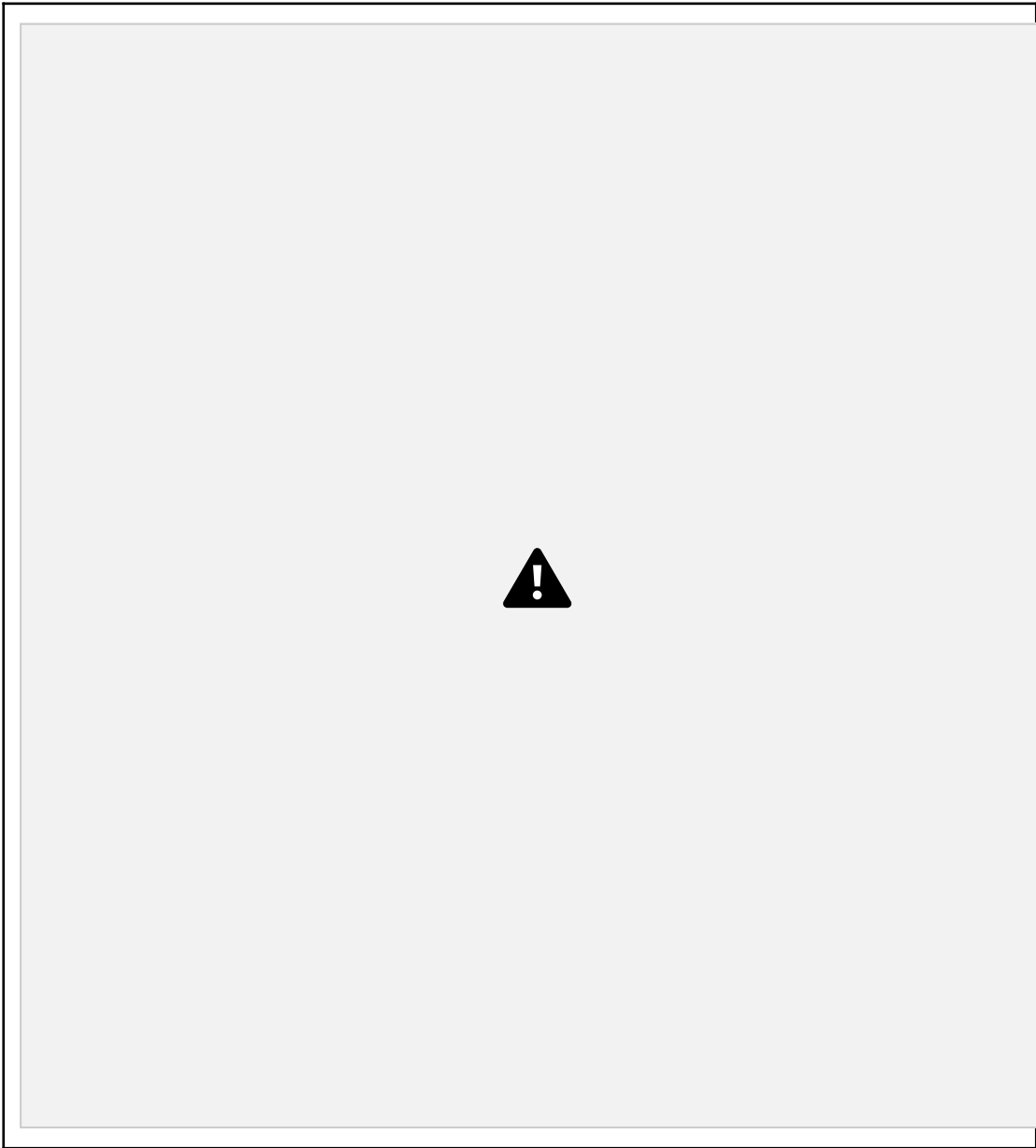


Figure 3.31: Use Case Diagram for CyberSafe Learning Hub

The Use Case Model in Figure 3.31 illustrates the interactions between the CyberSafe Learning Hub system and its two primary actors: the Student and the Administrator (Lecturer). It outlines the key operations available to each actor, defines the boundaries of the system, and serves as a reference for understanding how each user type engages with the platform to fulfil their

respective goals.

The Student is the primary end user of the system. A student may register for a new account using their email and password or sign in through Google OAuth 2.0. Upon logging in for the first time, the student is guided through a brief onboarding process before being directed to the Student Dashboard. From the dashboard, the student can browse available cybersecurity learning modules organised by category, view individual lessons, and mark lessons as complete to track their progress. Upon completing a lesson, the student may attempt the associated quiz.

The quiz system enforces an 85% passing threshold, and students who do not meet this requirement may retake the quiz. During a quiz, the student may interact with the built-in AI Chatbot, which provides concept explanations without revealing direct answers. Upon passing all quizzes within a module, the student is automatically awarded a module badge. When all modules within a category are completed, the system generates and issues a downloadable PDF certificate. The student may also view the Leaderboard, which ranks all students based on their performance scores, encouraging healthy competition and sustained engagement.

The Administrator is responsible for managing all platform content and monitoring student activity. Although the administrator uses the same login entry point as the student, the system detects their role and redirects them to the Admin Dashboard upon authentication. From the admin side, the administrator can create, edit, and delete categories, modules, lessons, and lesson content blocks, as well as manage quizzes and their associated questions and answers. The administrator can also manage registered user accounts, including updating roles and removing users from the system. Through the Analytics module, the administrator can monitor individual student performance, view quiz attempt histories, and track overall platform progress. Additionally, the administrator may send in-platform notifications to individual students or broadcast announcements to all students.

The Use Case Model clearly distinguishes the roles and responsibilities of both actors while highlighting their shared entry point through the authentication process. It provides a structured view of the system's functional scope and confirms that the CyberSafe Learning Hub is designed to support both effective content delivery for students and efficient platform management for administrators.

3.5.3 Flowchart

3.5.3.1 Student Flowchart

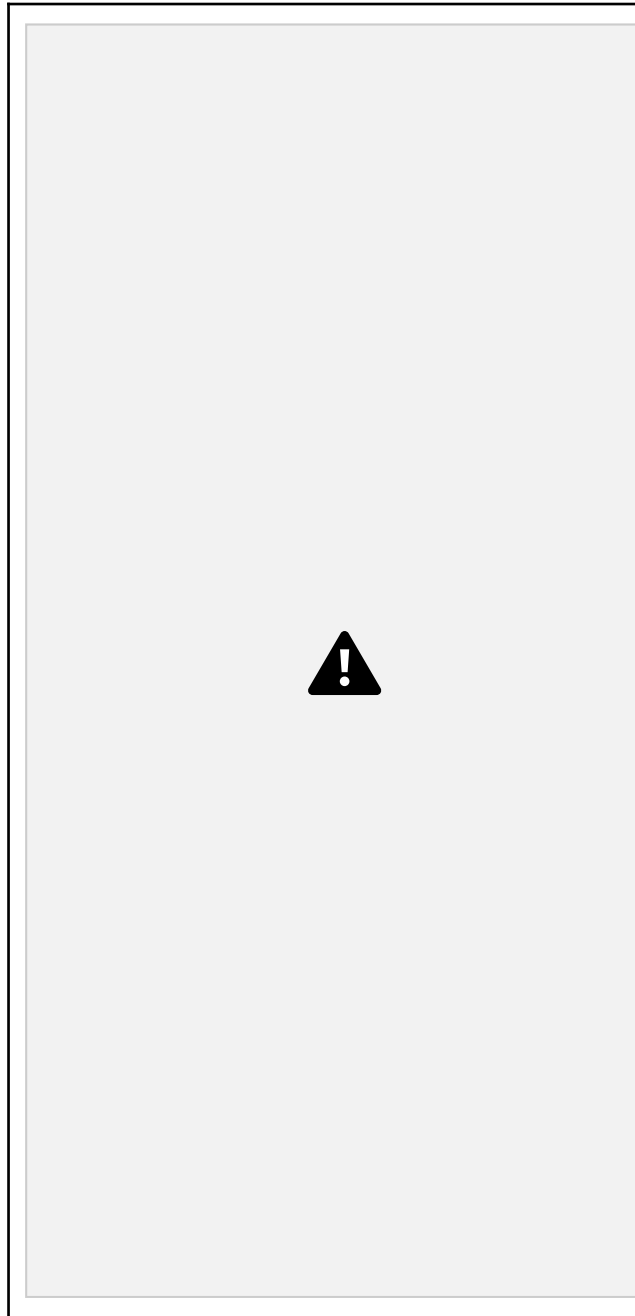


Figure 3.32: User (Student) Flowchart of CyberSafe Learning Hub

Figure 3.32 illustrates the process a student goes through when using the CyberSafe Learning Hub. The flow begins when the student accesses the system and chooses to either register a new account or log in using their email and password or Google OAuth 2.0. If authentication fails, an error message is displayed and the student is prompted to try again. Upon successful authentication, the student is directed to the Student Dashboard, where they can browse and select a cybersecurity learning module. The student proceeds to view the lesson content and, upon completion, attempts the quiz associated with the lesson. The system evaluates the submission and checks whether the student has achieved a score of 85% or above. If the score is below the passing threshold, the

student may retake the quiz. Once the quiz is passed, the system automatically awards a badge or certificate based on the student's progress. The student is then prompted to continue learning or log

out. If they choose to continue, they are redirected to the dashboard to select the next module. The flow concludes when the student logs out of the system.

3.5.3.2 Admin Flowchart

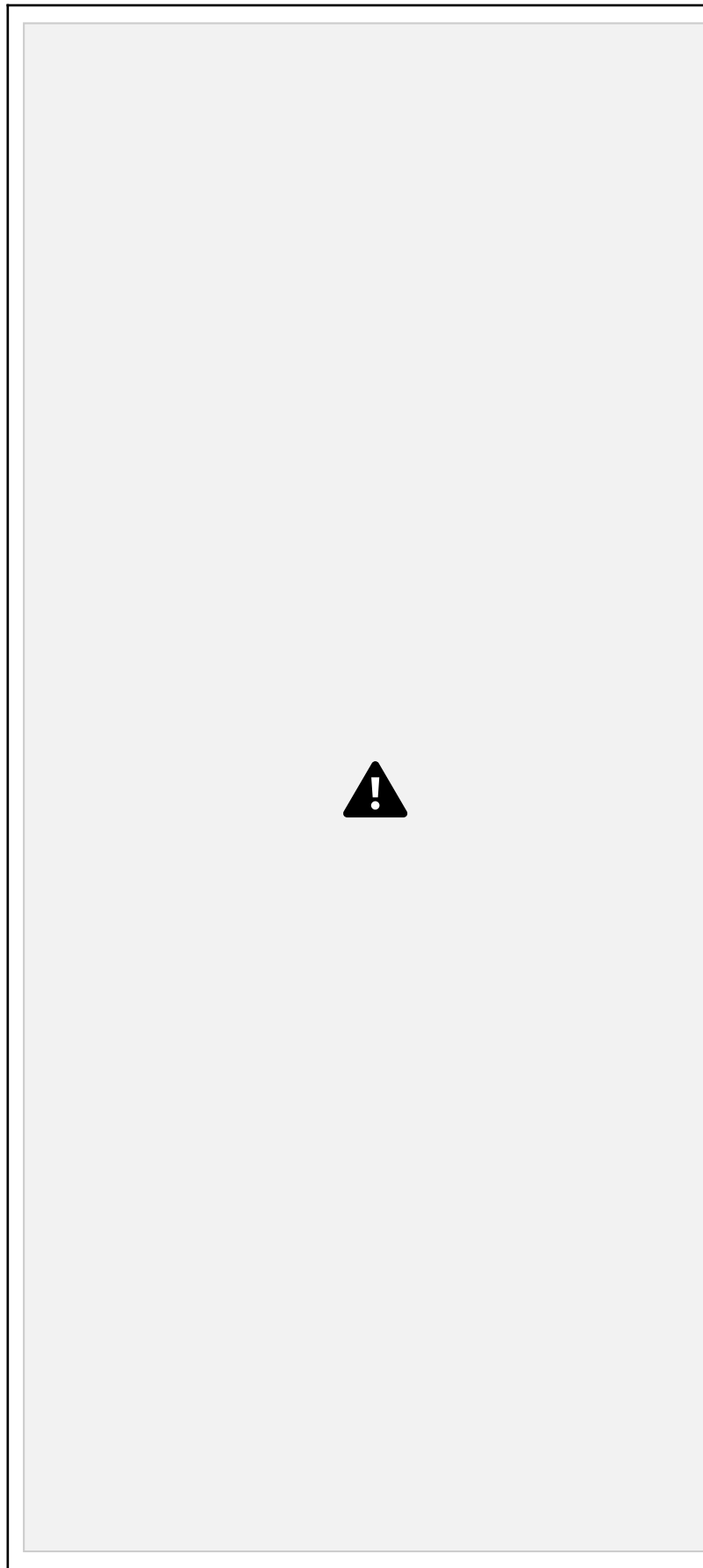


Figure 3.33: User (Administrator/Lecturer) Flowchart of CyberSafe Learning Hub

Figure 3.33 depicts the process an administrator goes through when using the CyberSafe

Learning Hub. The flow begins when the administrator navigates to the login page and enters their credentials. If authentication fails, an error message is displayed and the administrator is prompted to try again. Upon successful authentication, the administrator is directed to the Admin Dashboard, which provides an overview of platform statistics and activity. From the dashboard, the administrator can perform a series of management tasks: managing content such as categories, modules, lessons, and quizzes; managing user accounts including editing roles and removing users; viewing analytics to monitor student performance and progress; and sending notifications to individual students or all students on the platform. After completing the required tasks, the administrator is asked whether they wish to continue using the system. If yes, they are returned to the Admin Dashboard. If no, the administrator logs out and the process ends.

3.6 Conclusion

In summary, this chapter explained the process used to create CyberSafe Learning Hub as a learning environment for increasing cybersecurity awareness and literacy in college students. The chapter explained how Agile methodology was utilized to enable iterative development, embed user feedback and continuous improvement within the project. Both quantitative and qualitative methods are applied in order to make the system design consistent with real user needs and learning behaviours.

The information from the questionnaires and interviews helped to inform us about what users knew or did not know regarding cybersecurity, the system for learning that they preferred, system expectation and so forth. According to the questionnaire, although students were aware of the importance of cybersecurity, many did not have a realistic appreciation nor did they adopt safe online behaviour to protect themselves. Interview with the lecturer provided insights from an educator's perspective as well confirming the requirement for more structured, interactive and engaging digital learning materials

The synthesis of this evidence informed the design decisions for the system's functional and non functional requirements such as gamification, user authentication, progress tracking, usability and performance. These needs provided the basis for the consequent project design and development phases.

In conclusion, this chapter presented the methodological platform upon which rests the whole project. By building the system upon empirical evidence and stakeholder co-creation, the CyberSafe Learning Hub is structured to be both technically secure and robust as well as pedagogically effective and engaging in raising awareness among students about cybersecurity.

4 DESIGN & IMPLEMENTATION

4.1 Introduction

This chapter includes the general layout of the proposed system. The design phase is concerned with converting the system requirements into a detailed and structured blue print that is used to develop the system. It discusses the interface design, database design, system architecture, and security issues so that the system can be functional, reliable and secure. The interaction between the system components and their interaction with each other and the way the system is going to interact with the users is depicted using diagrams and design models. The chapter is a source of implementation phase that is discussed in the next chapter.

4.2 Interface Design

The interface design section shows the wireframes that have been designed in the CyberSafe Learning Hub before the implementation. The wireframes are organised into three parts. The first section deals with the general pages which includes the Landing Page, Login Page, and Register Page that is publicly available to all users irrespective of their roles. The second and third sections discuss the role-specific interfaces of the administrators and students respectively.

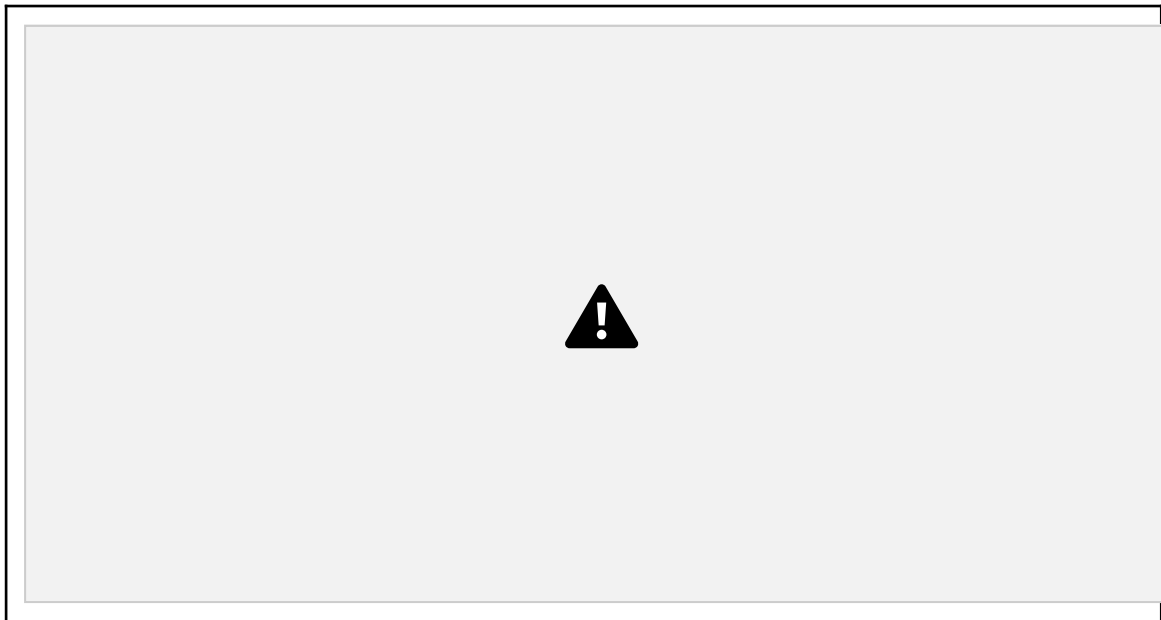


Figure 4.1: Landing Page

Figure 4.1 represents the opening page of the CyberSafe Learning Hub. It is the point of entry into the system and it presents the users with the platform through the CyberSafe logo and a welcome message. The page emphasizes essential elements of the visuals with image cards and offers two major action buttons, Sign In and Register, where users can enter the existing accounts or create a new one.

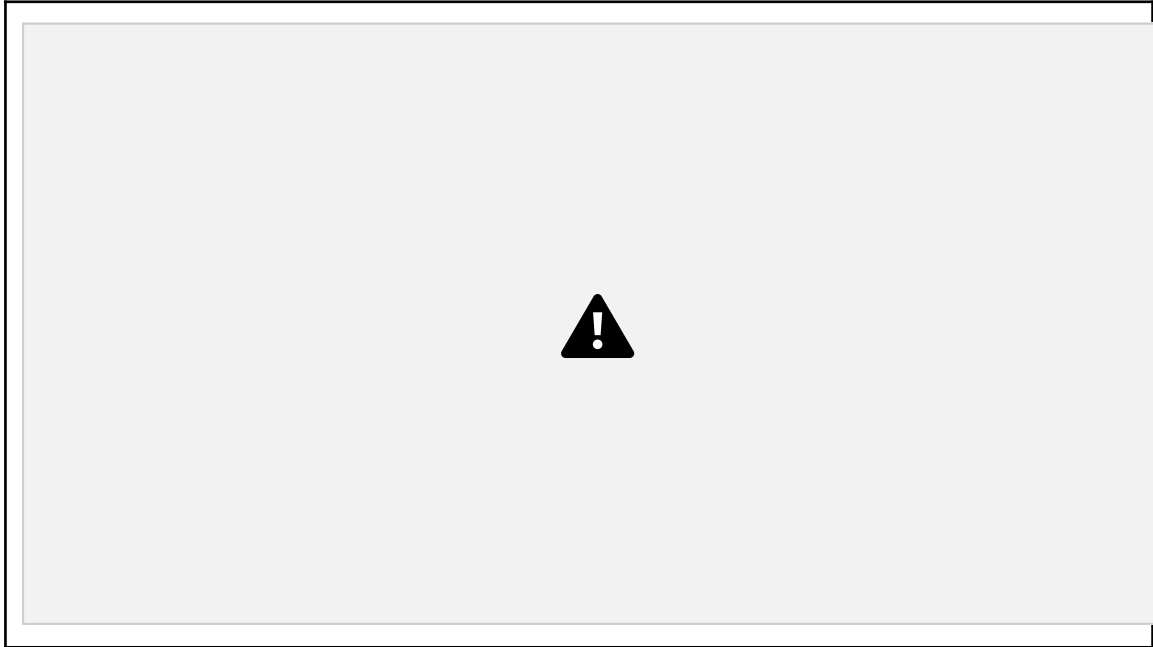


Figure 4.2: Login Page

The system has a login page as illustrated in figure 4.2. Registered users can use this page to log in by typing in their email address and password. It also offers an alternative of Google sign-in to log in, which makes it convenient and accessible to users.

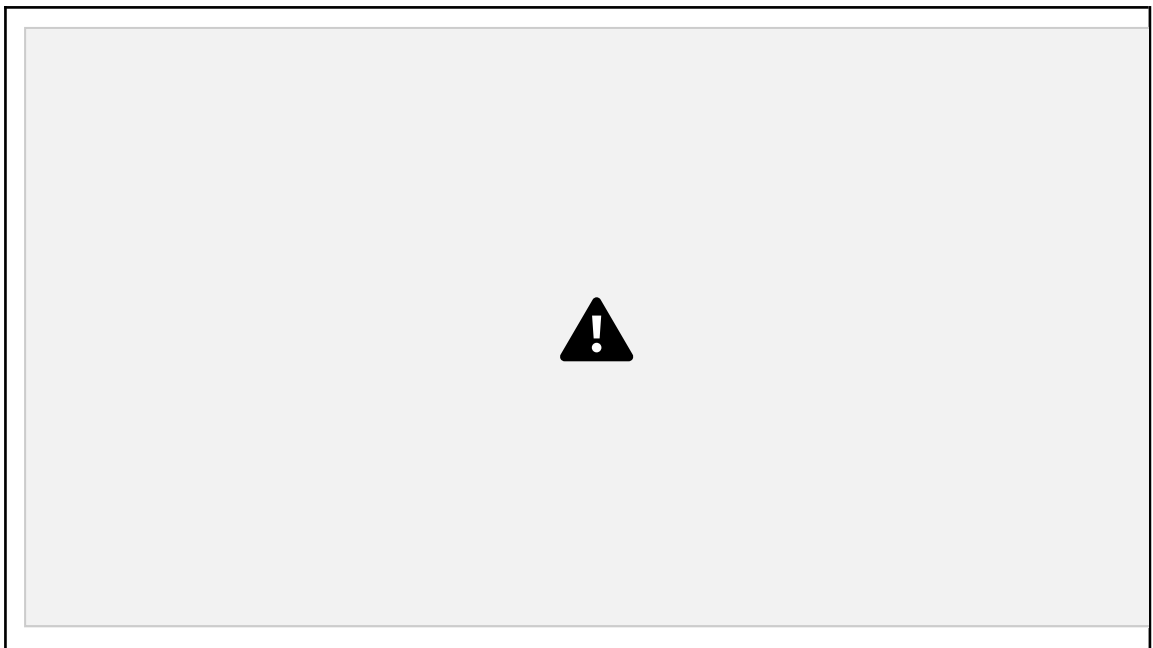


Figure 4.3: Register Page

Figure 4.3 indicates the registration page whereby new users are able to open an account. The page has textboxes where name, email address, password and password confirmation can be entered. Just like the login page, it has also provided a Google sign-in facility to make the registration process easier.

4.2.1 Admin

The wireframes below represent the administrative side of the CyberSafe Learning Hub. These pages can be accessed by the user only under the administrator role and discuss the fundamental management capabilities of the platform such as dashboards, module and lesson management, quiz management, analytics, and user management.



Figure 4.4: Admin Dashboard Page

The page of the admin dashboard as depicted in figure 4.4 gives an overview of the system analytics. It shows major statistics including the total number of students, the total number of quizzes, number of quiz attempts, and average marks. Other features of the dashboard are module completion progress bars, top performers, quick statistics and a recent activity table to assist the administrators in monitoring the general performance of the platform.

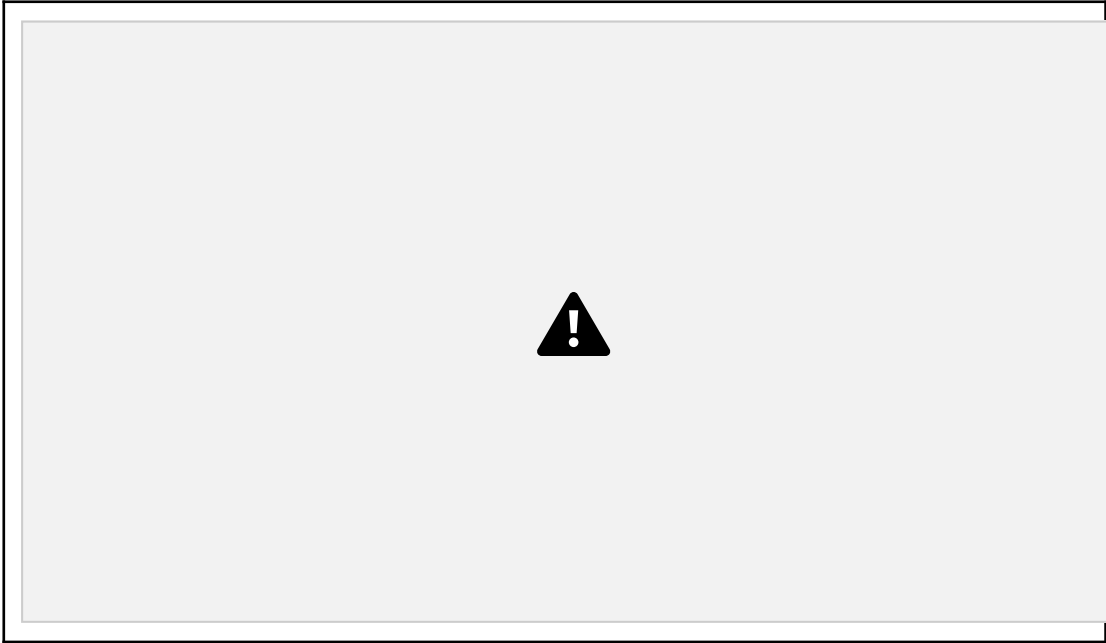


Figure 4.5: Manage Module Page

The figure 4.5 demonstrates the manage module page where administrators can see all the available modules in a card format. All the module cards have a module title, description and image with action buttons to edit or delete the modules. There is also a Create Module button where one can add new modules.

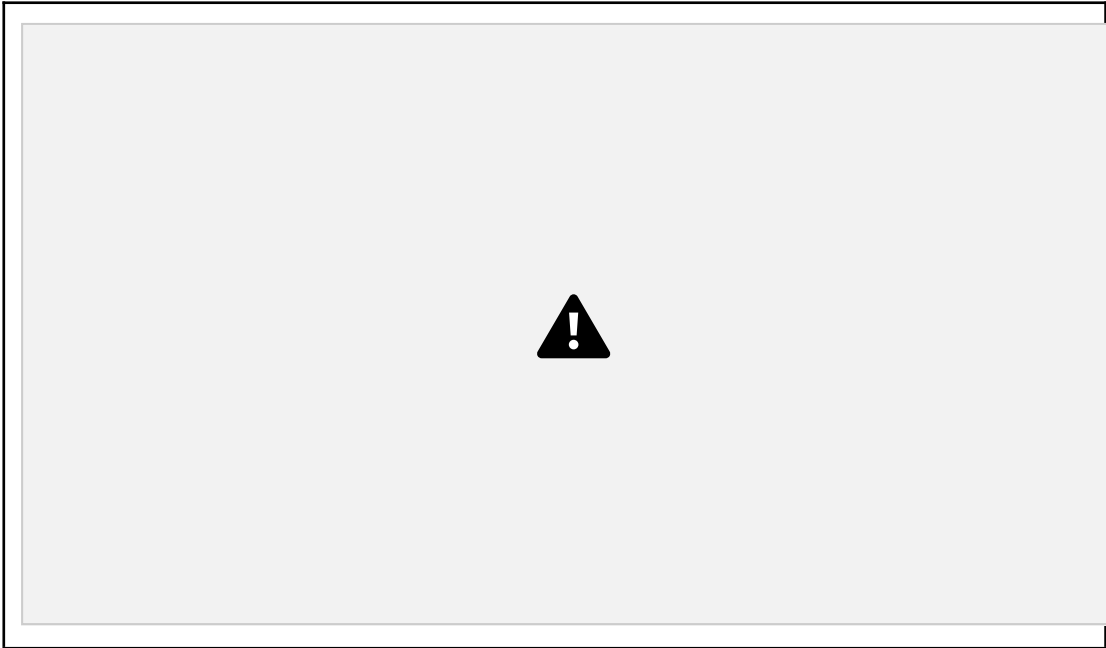


Figure 4.6: Manage Lessons Page

The manage lessons page is depicted in figure 4.6 and it enables the administrators to arrange and manage lessons within every module. The lessons are presented in a tabular format with the

lesson order, lesson title, quiz availability, and lesson editing and deletion action buttons. Lesson lists in each module are expanded or collapsed in dropdown menus.

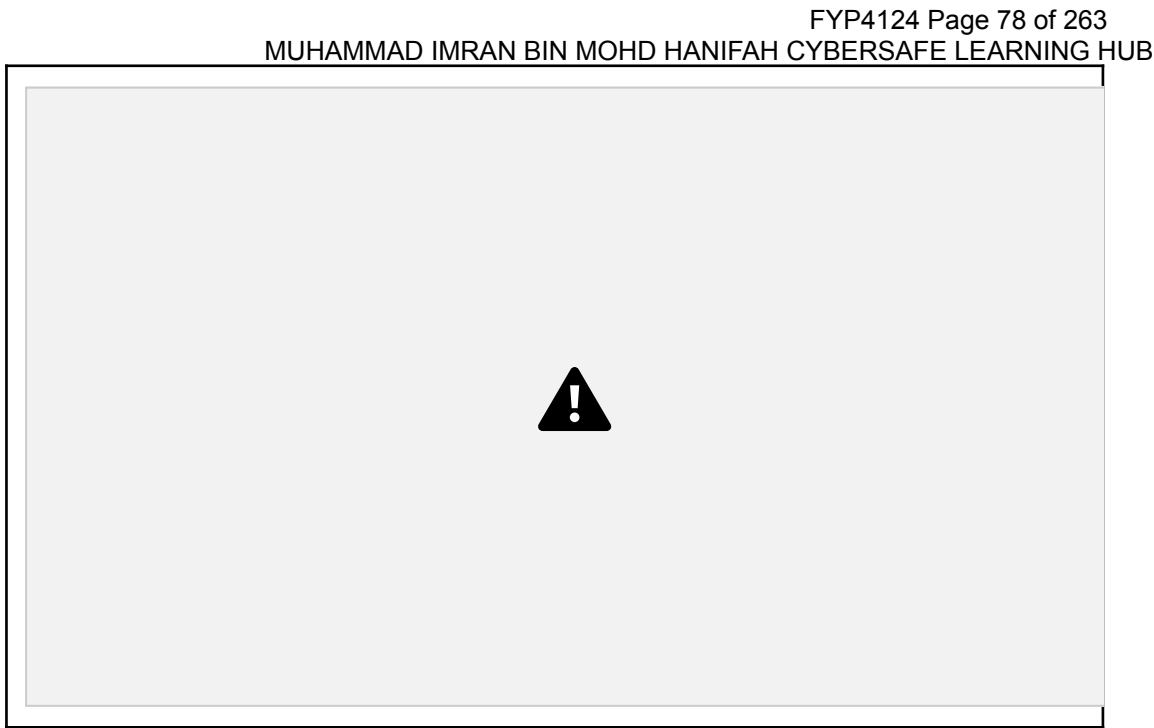


Figure 4.7: Manage Quizzes Page

The manage quizzes page is displayed in figure 4.7. This page shows quizzes in a tabular format with information about quiz title, the module and lesson to which the quiz was associated, the number of questions, attempts, and pass rate. Action buttons allow administrators to edit or delete quizzes easily.



Figure 4.8: Student Analytics Page

The student analytics page is illustrated in figure 4.8 and is dedicated to the analysis of student performance and learning progress. It offers some insights in the form of statistics, performance overviews, and activity tracking, which helps administrators to assess the learning outcomes and determine the areas where improvements can be made.

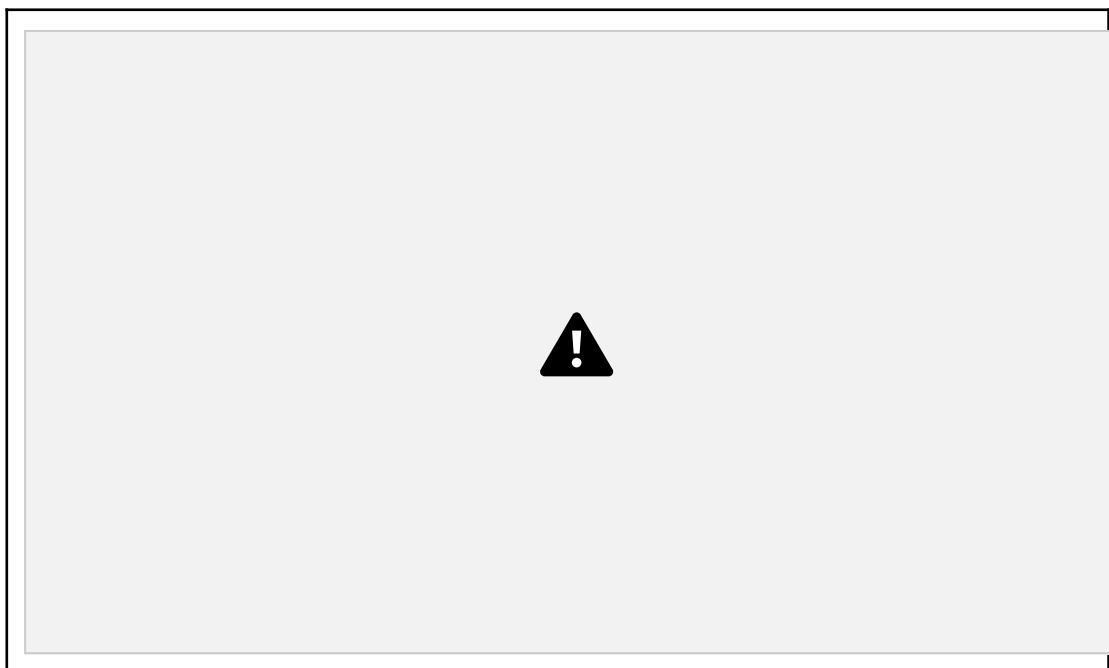


Figure 4.9: Manage Users Page

Figure 4.9 shows the manage users page, where administrators can manage system users. The table has the user names, email addresses, roles, date of registration and actions available. The administrators are able to change the user roles with dropdown menus and delete users with delete buttons which is effective in managing the users and controlling access.